



**Mplify Standard**  
**Mplify 99.1**

**LSO Service Ordering Management API**  
**Developer Guide**

**November 2025**

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# List of Contributing Members

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The following members of Mplify participated in the development of this document and have requested to be included in this list.

| Member                   |
|--------------------------|
| Amartus                  |
| Colt Technology Services |
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**Table 1. Contributing Members**

# 1. Abstract

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This standard is intended to assist the implementation of the Application Programming Interfaces (APIs) for the Service Provisioning function of the Service Orchestration Functionality at the LSO Allegro, LSO Interlude and LSO Legato Interface Reference Points. The Interface Reference Points are defined in the MEF 55.1 [[MEF55.1](#)] at the interface between the Business Application Systems layer and Service Orchestration Functionality layer.

This standard normatively incorporates the following files by reference as if they were part of this document from the GitHub repository:

## MEF-LSO-Allegro-SDK

commit id: [e79d2e77dc818aa913f8b66e9108c4ff4e2f1297](#)

- [serviceApi/order/serviceOrderingManagement.api.yaml](#)
- [serviceApi/order/serviceOrderingNotification.api.yaml](#)

## MEF-LSO-Interlude-SDK

commit id: [dda56a69cf22a63660c0e03a726c310cd3b29dbc](#)

- [serviceApi/order/serviceOrderingManagement.api.yaml](#)
- [serviceApi/order/serviceOrderingNotification.api.yaml](#)

## MEF-LSO-Legato-SDK

commit id: [f4bc5595fb5283d3c30a485099a9d12ba29757ee](#)

- [serviceApi/order/serviceOrderingManagement.api.yaml](#)
- [serviceApi/order/serviceOrderingNotification.api.yaml](#)

## 2. Terminology and Abbreviations

This section defines the terms used in this document. In many cases, the normative definitions to terms are found in other documents. In these cases, the third column is used to provide the reference that is controlling, in other Mplify or external documents.

In addition, terms defined in the following documents are included in this document by reference, and are not repeated in the tables below.

- [MEF 55.1](#)
- [MEF 55.1.1](#)
- [Mplify 150](#)

| Term                                | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                      | Source                                          |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|
| API Endpoint                        | The endpoint of a communication channel (the complete URL of an API Resource) to which the HTTP-REST requests are addressed in order to operate on the <i>API Resource</i>                                                                                                                                                                                                                                                                      | <a href="#">rapidapi.com</a><br>This document   |
| API Resource                        | A REST Resource. In REST, the primary data representation is called Resource. In this document, <i>API Resource</i> is defined as a OAS <i>SchemaObject</i> with specified <i>API Endpoints</i>                                                                                                                                                                                                                                                 | <a href="#">restfulapi.net</a><br>This document |
| Business Applications               | The Service Provider functionality supporting Business Management Layer functionality                                                                                                                                                                                                                                                                                                                                                           | MEF 55.1                                        |
| OAS Document                        | An API description document in the OpenAPI specification format.                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">openapis.org</a>                    |
| OpenAPI                             | The OpenAPI 3.0 Specification, formerly known as the Swagger specification is an API description format for REST APIs.                                                                                                                                                                                                                                                                                                                          | <a href="#">spec.openapis.org</a>               |
| Operation                           | An interaction between the BUS and SOF, potentially involving multiple back and forth transactions.                                                                                                                                                                                                                                                                                                                                             | This document                                   |
| SchemaObject                        | The construct that allows the definition of input and output data types. These types can represent object classes, as well as primitives and arrays. specification                                                                                                                                                                                                                                                                              | <a href="#">spec.openapis.org</a>               |
| Service Orchestration Functionality | The set of service management layer functionality supporting an agile framework to streamline and automate the service lifecycle in a sustainable fashion for coordinated management supporting design, fulfillment, control, testing, problem management, quality management, usage measurements, security management, analytics, and policy-based management capabilities providing coordinated end-to-end management and control of Services | MEF 55.1                                        |

**Table 2. Terminology**

| Term | Definition                                                                                   | Source        |
|------|----------------------------------------------------------------------------------------------|---------------|
| API  | Application Programming Interface. In this document, API is used synonymously with REST API. | This document |
| BUS  | Business Applications                                                                        | MEF 55.1      |

| <b>Term</b> | <b>Definition</b>                   | <b>Source</b>                                   |
|-------------|-------------------------------------|-------------------------------------------------|
| IRP         | Interface Reference Point           | This document                                   |
| OAS         | OpenAPI Specification               | <a href="https://openapis.org">openapis.org</a> |
| SOF         | Service Orchestration Functionality | MEF 55.1                                        |

**Table 3. Abbreviations**



### 3. Compliance Levels

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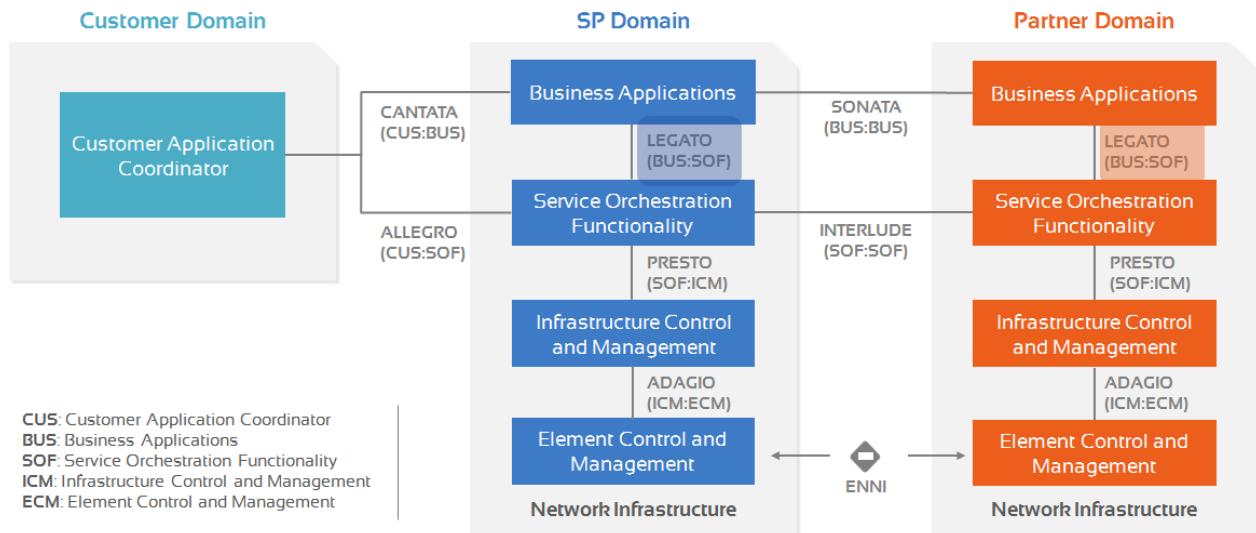
The key words **"MUST"**, **"MUST NOT"**, **"REQUIRED"**, **"SHALL"**, **"SHALL NOT"**, **"SHOULD"**, **"SHOULD NOT"**, **"RECOMMENDED"**, **"NOT RECOMMENDED"**, **"MAY"**, and **"OPTIONAL"** in this document are to be interpreted as described in BCP 14 ([RFC 2119], [RFC 8174]) when, and only when, they appear in all capitals, as shown here. All key words must be in bold text.

Items that are **REQUIRED** (contain the words **MUST** or **MUST NOT**) are labeled as **[Rx]** for required. Items that are **RECOMMENDED** (contain the words **SHOULD** or **SHOULD NOT**) are labeled as **[Dx]** for desirable. Items that are **OPTIONAL** (contain the words **MAY** or **OPTIONAL**) are labeled as **[Ox]** for optional.

A paragraph preceded by **[CRa]<** specifies a conditional mandatory requirement that **MUST** be followed if the condition(s) following the "<" have been met. For example, **"[CR1]<[D38]"** indicates that Conditional Mandatory Requirement 1 must be followed if Desirable Requirement 38 has been met. A paragraph preceded by **[Cdb]<** specifies a Conditional Desirable Requirement that **SHOULD** be followed if the condition(s) following the "<" have been met. A paragraph preceded by **\*\*[COc]<\*\*** specifies a Conditional Optional Requirement that **MAY** be followed if the condition(s) following the "<" have been met.

## 4. Introduction

This standard specification document describes the Application Programming Interface (API) for Service Order Management functionality of the LSO Allegro, LSO Interlude and LSO Legato Interface Reference Points (IRP). The LSO Reference Architecture is shown in Figure 1 with the IRP highlighted.



**Figure 1. The LSO Reference Architecture**

Within the Legato IRP the API Client - Server are BUS - SOF, respectively. Within Allegro and Interlude - both the Client and the Server are SOF, so they are also called Buyer/Seller. In this revision of document term BUS is used to refer to the Client side and the term SOF is used to refer to the Server side of the API.

### 4.1. Description

This standard is scoped to cover APIs for following Service Orchestration Functionalities:

- Service Ordering and Fulfillment
  - Includes Service Configuration & Activation functions
- Service Notification
  - Includes Event Subscription/Hub and Listener notification functions

Other Service Orchestration Functionalities not addressed in this standard include (but not limited to):

- Service Inventory Management
- Service Catalog Management
- Service Qualification
- Service Activation Testing
- Service Problem Management
- Service Quality Management
- Service Usage measurements and Reporting (in support of billing)
- License Management

### 4.2. Conventions in the Document

- Code samples are formatted using code blocks. When notation `<< some text >>` is used in the payload sample it indicates that a comment is provided instead of an example value and it

- might not comply with the OpenAPI definition.
- Model definitions are formatted as in-line code (e.g. `ServiceOrder`).
- In UML diagrams the default cardinality of associations is `0..1`. Other cardinality markers are compliant with the UML standard.
- In the API details tables and UML diagrams required attributes are marked with a `*` next to their names.
- In UML sequence diagrams `{{variable}}` notation is used to indicate a variable to be substituted with a correct value.

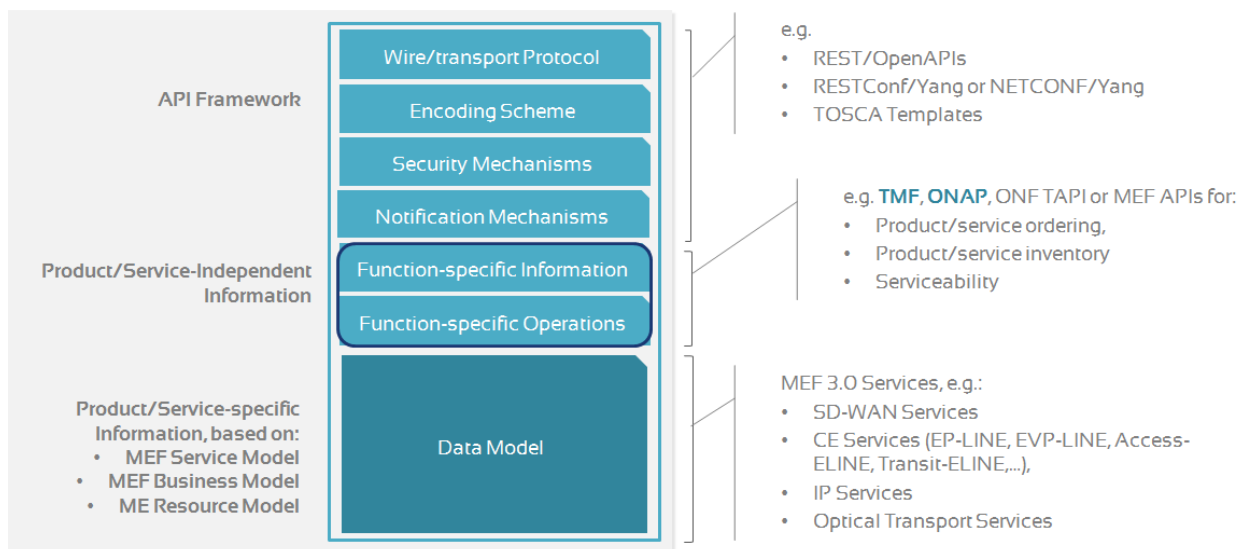
### 4.3. Relation to Other Documents

The API definition builds on *TMF641 Service Order Management API REST Specification v4.1.0* [TMF641]. Service Order Use Cases must support the use of any of Mplify service specifications as payload, in particular those defined in:

### 4.4. Approach

As presented in Figure 2, the API framework consist of three structural components:

- Generic API framework
- Service-independent information (Function-specific information and Function-specific operations)
- Service-specific information (Mplify service specification data model)



**Figure 2. API Structure**

The essential concept behind the framework is to decouple the common structure, information, and operations from the specific service information content.

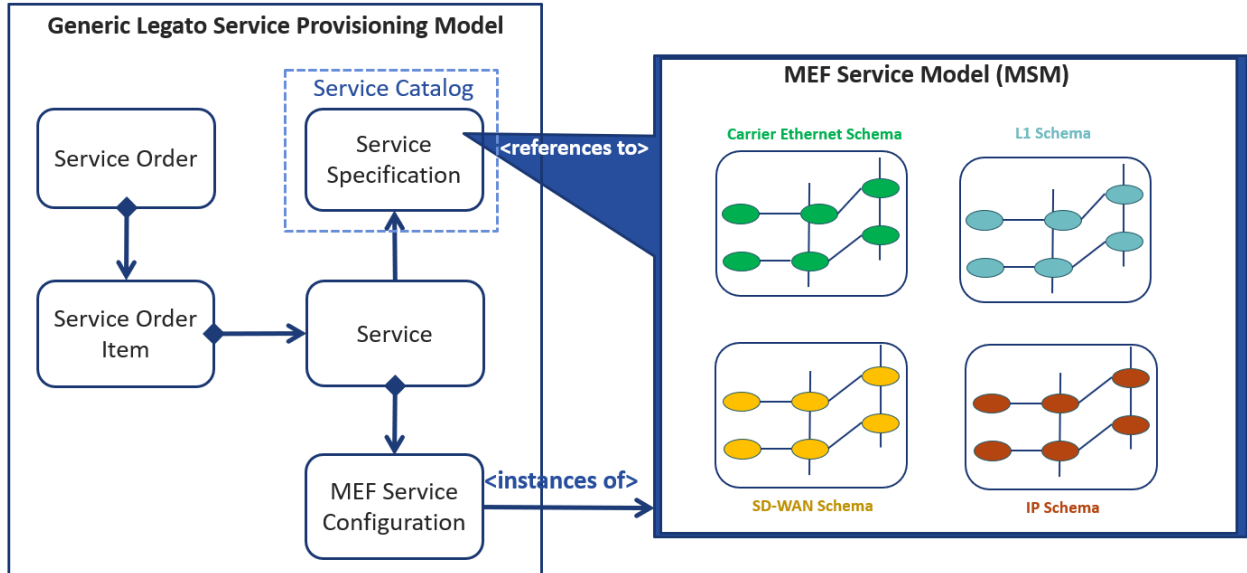
Firstly, the Generic API Framework defines a set of design rules and patterns that are applied across all APIs.

Secondly, the service-independent information of the framework focuses on a model of a particular functionality and is agnostic to any of the service specifications. For example, this standard is describing the Service Order model and operations that allow ordering of any service that is aligned with either Mplify or custom service specifications.

Finally, the service-specific information part of the framework focuses on Mplify service specifications that define business-relevant attributes and requirements for trading Mplify subscriber and Mplify operator services.

This Developer Guide is not defining Mplify service specifications but can be used in combination with any service specifications defined by or compliant with Mplify.

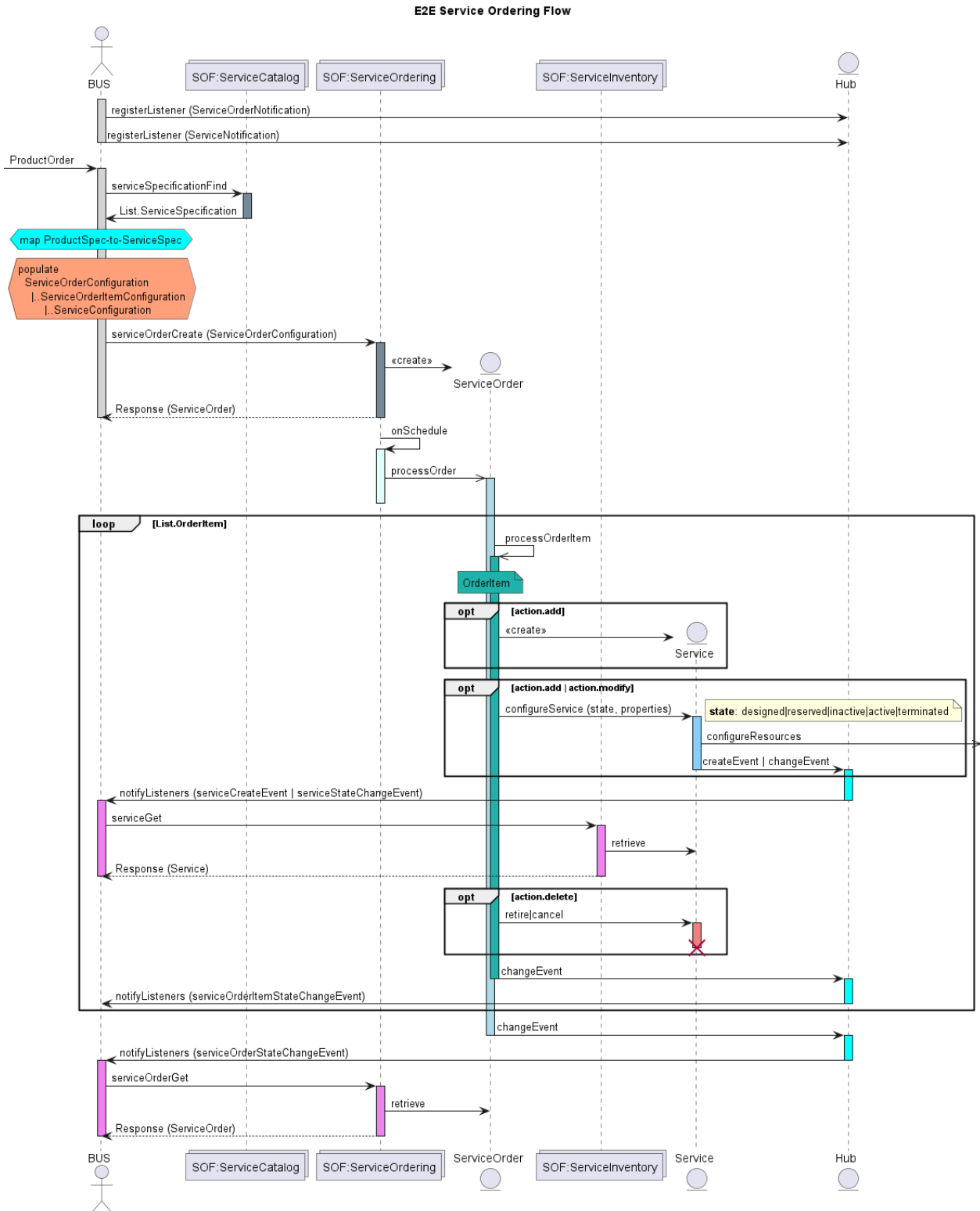
Figure 3 presents the relations between the API components and the Service Model. A Service Order contains one or more Service Order Items. Each Service Order Item is an intent of action on a given Service (**add**, **modify** or **delete**). A Service references Service Specification to identify the Service Type. The Service specification points to the schema of the Service, as provided by (but not limited to) Mplify Standard. The Service also has the **MefServiceConfiguration** attribute, which provides an instance of the configuration of a given Service (attributes of Mplify Service model populated with desired values)



**Figure 3. MSM Schema**

#### 4.5. High-Level Flow

The Service Catalog, Service Order, Service Inventory, and Service Notification APIs in essence allow the BUS to request SOF to configure and activate one or more services as part of an order fulfillment process. Figure 4 presents a high-level flow of use of all of the above-mentioned APIs.



**Figure 4. High-Level Flow**

The following steps describe the high-level flow:

- The BUS system registers for notifications.
- As part of the ordering flow, the BUS system receives the product order (through Cantata or Sonata) which triggers the fulfillment processes in the BUS system.
- The BUS system first queries the *Service Catalog* to retrieve the *ServiceSpecifications* supported by the SOF

**Note1:** *Service Catalog and the process of mapping and decomposing a product order to identify appropriate *ServiceSpecifications* is out of scope for this standard.* **Note2:** *The*

*mechanisms to design, construct and populate the **ServiceSpecifications** into SOF Service Catalog is out of scope for this standard.*

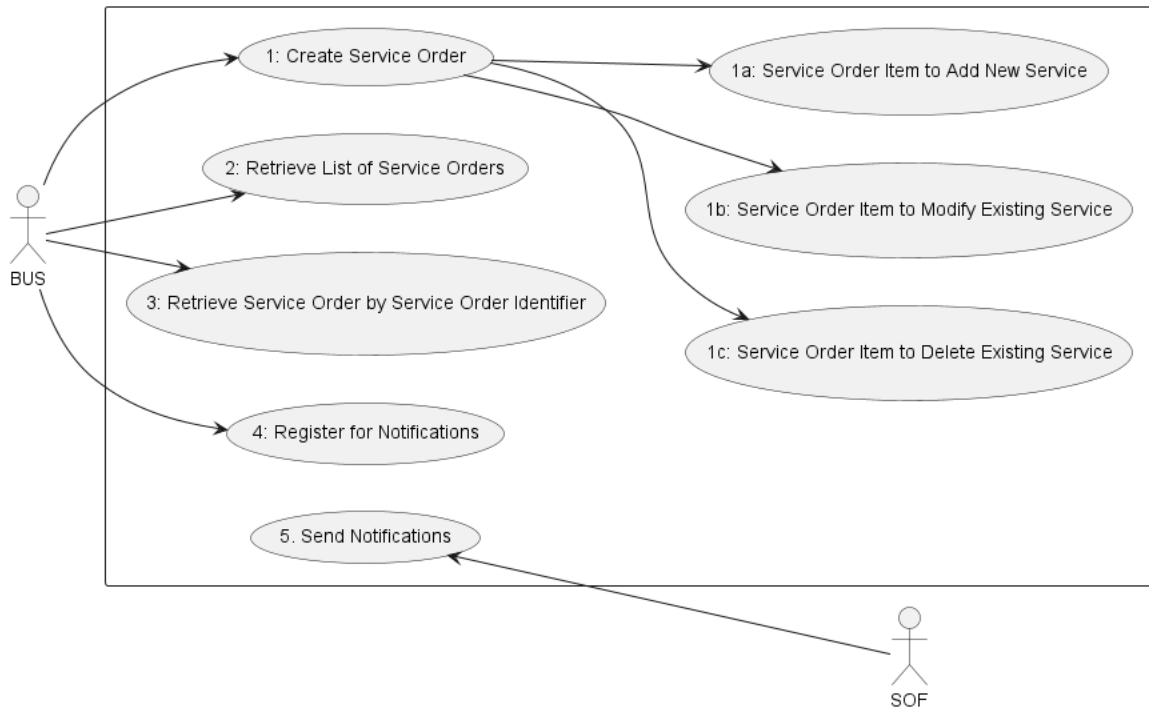
- Each specific instance of a **ServiceSpecification** (retrieved from the *Service Catalog*) minimally contains a reference to target **Service** schema. A Service schema describes the set of properties that characterize that service and are exchanged.
- During the service configuration and activation phase, the BUS system uses the *Service Order API* to instantiate the **Service** utilizing the **ServiceSpecifications** (retrieved from the *Service Catalog*).
  - The BUS achieves this by creating a **ServiceOrder** which contains a one or more **ServiceOrderItems**.
  - Each **ServiceOrderItem** carries some **ServiceConfiguration** data and the type of operation (*add/modify/delete*) to be performed (instructions to SOF).
  - The SOF utilizes **Service** schema referenced in the **ServiceSpecification** to validate the **ServiceConfiguration** data passed in by the BUS.
  - The **ServiceOrder** / **ServiceOrderItem** is processed by the SOF as per the state transition rules described in [6.1.7. Service Order and Service Order Items State Machine](#)
  - The SOF reports the **ServiceOrder** and **ServiceOrderItem** state changes
  - The SOF performs the actions (*add/modify/delete*) specified in a **ServiceOrderItem** on the specified target **Service** instance in the *Service Inventory* as per the state transition rules described in [6.6. Service Lifecycle](#)
  - The SOF reports the **Service** instance state changes
- The BUS system uses the same *Service Order API* to create **new Service** instances as well as update **existing Service** instance's properties or trigger state transitions, and disconnect **existing Service** instance.

## 5. API Description

This section presents the API structure and design patterns. It starts with the high-level use cases diagram. Then it describes the REST endpoints with use case mapping. Next, it gives an explanation of the design pattern that is used to combine service-agnostic and service-specific parts of API payloads. Finally, payload validation and API security aspects are discussed.

### 5.1. High-level Use Cases

Figure 5. presents a high-level use case diagram. It aims to help understand the endpoint mapping. Use cases are described extensively in [chapter 6](#)



**Figure 5.** Use cases

### 5.2. API Endpoints and Operations Summary

#### 5.2.1. SOF Service Ordering API Endpoints

**Allegro - Base URL:**

```
https://{{serverBase}}:{{port}}
{{?/sof_prefix}}/mefApi/allegro/serviceOrderingManagement/v1/
```

**Interlude - Base URL:**

```
https://{{serverBase}}:{{port}}
{{?/sof_prefix}}/mefApi/interlude/serviceOrderingManagement/v1/
```

**Legato - Base URL:**

```
https://{{serverBase}}:{{port}}
{{?/sof_prefix}}/mefApi/legato/serviceOrderingManagement/v6/
```

The following API Endpoints are used by BUS to create and query for **ServiceOrder** instances and to subscribe/unsubscribe to **ServiceOrder** notifications. The endpoints and corresponding

data model are defined in `serviceApi/order/serviceOrderingManagement.api.yaml`

| API Endpoint             | Description                                                                                                                                                                                                                              | Use Case mapping                                         |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| POST /serviceOrder       | A request initiated by the BUS to <i>create</i> new <b>Service</b> instances as well as <i>update</i> <b>Service</b> instance's properties or trigger their state transitions and/or <i>disconnect</i> existing <b>Service</b> instance. | UC 1: Create Service Order                               |
| GET /serviceOrder        | A request initiated by the BUS to retrieve a list of <b>ServiceOrders</b> from the service order management system in SOF, that match the filter criteria provided as <i>query</i> parameters                                            | UC 2: Retrieve List of Service Orders                    |
| GET /serviceOrder/{{id}} | A request initiated by the BUS to retrieve a specific <b>ServiceOrder</b> from the service order management system in SOF, that match the <i>id</i> provided as <i>path</i> parameter                                                    | UC 3: Retrieve Service Order by Service Order Identifier |
| POST /hub                | A request initiated by the BUS to instruct the SOF to send notification                                                                                                                                                                  | UC 4: Register for Notifications                         |
| GET /hub/{{id}}          | A request initiated by the BUS to retrieve a specific <b>EventSubscription</b> from the service order management system in SOF, that matches the provided <i>id</i> provided as <i>path</i> parameter                                    | UC 4: Register for Notifications                         |
| DELETE /hub/{{id}}       | A request initiated by the BUS to instruct the SOF to stop sending notifications                                                                                                                                                         | UC 4: Register for Notifications                         |

**Table 4. SOF Service Ordering API Endpoints**

[R1] SOF **MUST** support all API endpoints listed in Table 4.

### 5.2.2. BUS Service Ordering API Endpoints

**Allegro - Base URL:**

`https://{{serverBase}}:{{port}}  
{{?/sof_prefix}}/mefApi/allegro/serviceOrderingNotification/v1/`

**Interlude - Base URL:**

`https://{{serverBase}}:{{port}}  
{{?/sof_prefix}}/mefApi/interlude/serviceOrderingNotification/v1/`

**Legato - Base URL:**

`https://{{serverBase}}:{{port}}  
{{?/sof_prefix}}/mefApi/legato/serviceOrderingNotification/v6/`

The following API Endpoints are used by SOF to post notifications to registered BUS listeners. The endpoints and corresponding data model are defined in `serviceApi/order/serviceOrderingNotification.api.yaml`



| API Endpoint                                                     | Description                                                                                                                 | Use Case mapping      |
|------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|
| <code>POST /listener/serviceOrderCreateEvent</code>              | A request initiated by the SOF to notify BUS on <i>ServiceOrder</i> instance creation                                       | 5. Send Notifications |
| <code>POST /listener/serviceOrderInformationRequiredEvent</code> | A request initiated by the SOF to notify BUS that additional information is required for given <i>ServiceOrder</i> instance | 5. Send Notifications |
| <code>POST /listener/serviceOrderStateChangeEvent</code>         | A request initiated by the SOF to notify BUS on <i>ServiceOrder</i> instance state change                                   | 5. Send Notifications |
| <code>POST /listener/serviceOrderItemStateChangeEvent</code>     | A request initiated by the SOF to notify BUS on <i>ServiceOrderItem</i> instance state change                               | 5. Send Notifications |

**Table 5. BUS Service Ordering API Endpoints**

[O1] The BUS **MAY** support API endpoints listed in Table 5.

[O2] The BUS **MAY** register to receive service notifications.

[R2] The SOF **MUST** support sending notification to API endpoints listed in Table 5 to registered BUS.

### 5.3. Integration of Service Specifications into Service Order Management API

Service specifications are defined using JsonSchema (draft 7) format [JSON Schema draft 7](#) and are integrated into the *ServiceOrder* using the TMF extension pattern.

The extension hosting type in the API data model is *MefServiceConfiguration*. The `@type` attribute of that type must be set to a value that uniquely identifies the service specification. A unique identifier for Mplify standard service specifications is in URN format and is assigned by Mplify. This identifier is provided as root schema `$id` and in service specification documentation. Use of non-Mplify standard service definitions is allowed. In such a case the schema identifier must be agreed upon between the BUS and the SOF.

The example below shows a header of a Service Specification schema, which is describing the IP Uni, where `"$id": urn:mef:lso:spec:legato:ip-uni:v0.0.1:all` is the above-mentioned URN:

```
"$schema": http://json-schema.org/draft-07/schema#
"$id": $id": urn:mef:lso:spec:legato:ip-uni:v0.0.1:all
title: MEF LSO Legato - IP UNI Specification
```

Service specifications are provided as Json schemas without the *MefServiceConfiguration* context.

Service-specific attributes are introduced via the *ServiceValue* (defined by the BUS). This entity has the *serviceConfiguration* attribute of type *MefServiceConfiguration* which is used as an

extension point for service-specific attributes.

Implementations might choose to integrate selected service specifications to data model during development. In such a case an integrated data model is built and service specifications are in an inheritance relationship with `MefServiceConfiguration` as described in the OAS specification. This pattern is called **Static Binding**. The SDK is additionally shipped with a set of API definitions that statically bind all service-related APIs (POQ, Quote, Order, Inventory) with all corresponding service specifications available in the release. The snippets below present an example of a static binding of the envelope API with several Mplify service specifications, from both `MefServiceConfiguration` and service specification point of view:

```
MefServiceConfiguration:
  description:
    MefServiceConfiguration is used as an extension point for MEF-specific
    service payload. The '@type' attribute is used as a discriminator
  discriminator:
  mapping:
    urn:mef:lso:spec:legato:ip-enni:v0.0.1:all: '#/components/schemas/IpEnni'
    urn:mef:lso:spec:legato:ipvc-endpoint:v0.0.1:all: '#/components/schemas/IpvcEndpoint'
    urn:mef:lso:spec:legato:ip-uni:v0.0.1:all: '#/components/schemas/IpUni'
    urn:mef:lso:spec:legato:ethernet-uni-access-link-trunk:0.0.1:all:
    '#/components/schemas/EthernetUniAccessLinkTrunk'
    urn:mef:lso:spec:legato:ip-uni-access-link:0.0.1:all: '#/components/schemas/IpUniAccessLink'
    urn:mef:lso:spec:legato:ipvc:v0.0.1:all: '#/components/schemas/Ipvc'
    urn:mef:lso:spec:legato:ip-uni-access-link-trunk:0.1:all: '#/components/schemas/IpUniAccessLinkTrunk'
    urn:mef:lso:spec:legato:ip-enni-link:v0.0.1:all: '#/components/schemas/IpEnniLink'
  propertyName: '@type'
  properties:
    '@type':
      description:
        The name of the type, defined in the JSON schema specified above, for
        the service that is the subject of the Request. The named type must be
        a subclass of MefServiceConfiguration.
      type: string
```

```
IpvcEndpoint:
  allOf:
    - $ref: '#/components/schemas/MefServiceConfiguration'
    - description:
        'An IPVC End Point is a logical entity at an EI, to which a subset of
        packets that traverse the EI is mapped. Reference MEF 61.1 Section 7.4
        IP Virtual Connections and IPVC End Points.'
```

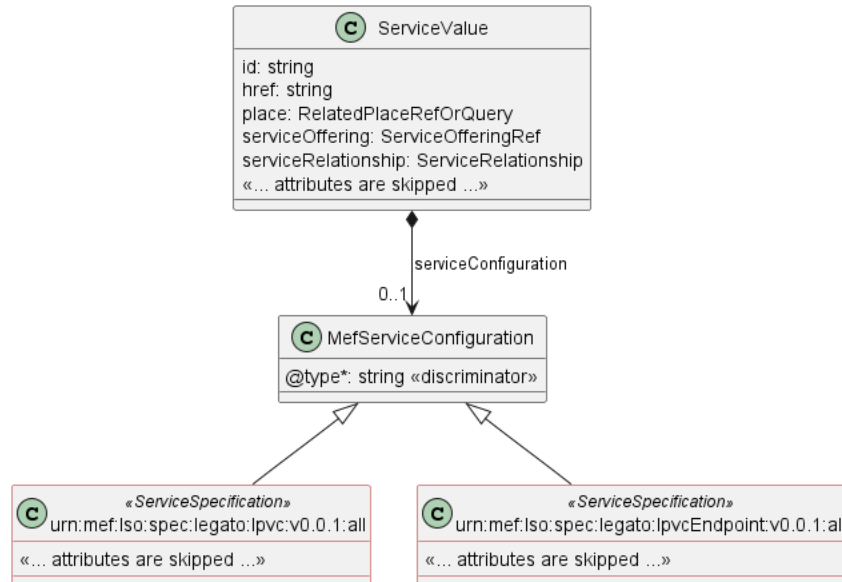
Alternatively, implementations might choose not to build an integrated model and choose a different mechanism allowing runtime validation of service-specific fragments of the payload. The system can validate a given service against a new schema without redeployment. This pattern is called **Dynamic Binding**.

Regardless of chosen implementation pattern, the HTTP payload is exactly the same. Both implementation approaches must conform to the requirements specified below.

[R3] `MefServiceConfiguration` type is an extension point that **MUST** be used to integrate service specifications' properties into a request/response payload.

[R4] The `@type` property of `MefServiceConfiguration` **MUST** be used to specify the type of the extending entity.

[R5] Service attributes specified in the payload must conform to the service specification specified in the `@type` property.



**Figure 6. The Extension Pattern with Sample Service-Specific Extensions**

Figure 6 presents two Mplify `<<ServiceSpecifications>>` that represent IPVC and IPVC Endpoint services. When these services are used as a Service Order payload the `@type` of `MefServiceConfiguration` takes `"urn:mef:lso:spec:legato:ipvc:v0.0.1:all"` or `"urn:mef:lso:spec:legato:ipvc-endpoint:v0.0.1:all"` value to indicate which service specification should be used to interpret a set of service-specific attributes included in the payload. An example of a service definition inside the `ServiceOrderItem` is presented in [Section 6.1.4](#).

The *all* suffix after the service type name in the URN comes from the approach that the service schemas may differ depending on the function (POQ, Quote, Order, or Inventory) they are used with. The value *all* means that one version of the schema is shared by all functions.

## 5.4. Sample Service Specification

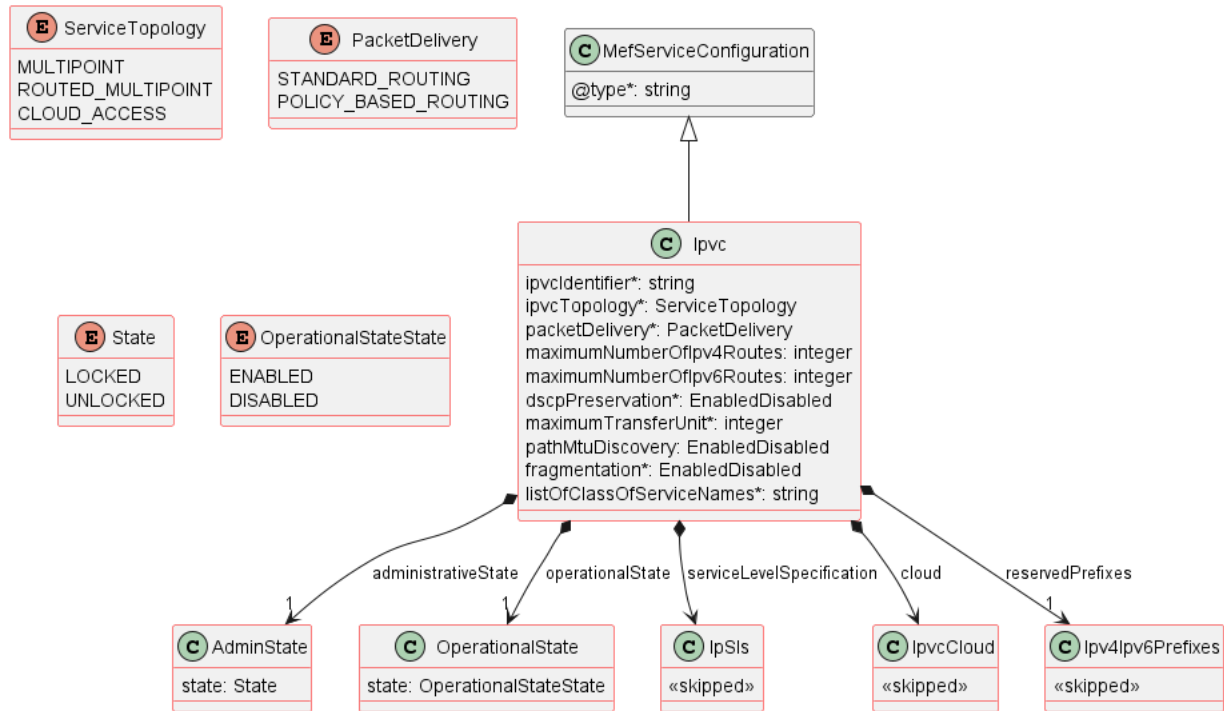
The SDK contains service specification definitions, from which IPVC and IPVC End Point are used in the payload samples in this section. The schemas are located in the SDK package at:

- `serviceSchema\ip\ipvc.yaml`
- `serviceSchema\ip\ipvcEndPoint.yaml`

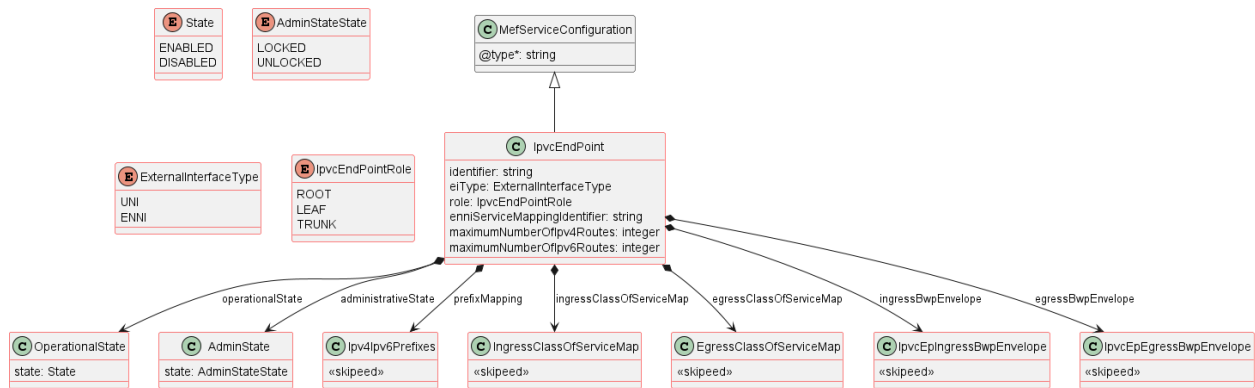
The service specification data model definitions are available as JsonSchema (version **draft 7**) documents. Figures 7 and 8 depict simplified UML views on these data models in which:

- the mandatory attributes are marked with **\***,
- the mandatory relations have a cardinality of **1** or **1..\***,
- some relations and attributes that are not essential to the understanding of the service specification model are omitted.

The red color in Figures 7 and 8 below highlights the data model of services. Some parts of the model are skipped for examples clarity. This is denoted by the `<<skipped>>` text in diagrams and in json snippets later in the document. Please note that this document uses service specifications just for the sake of example on how to use the Service Order API together with the Service payload. The detailed examples of any service specification are not in the scope of this document.



**Figure 7. A simplified view of IPVC service specification data model**

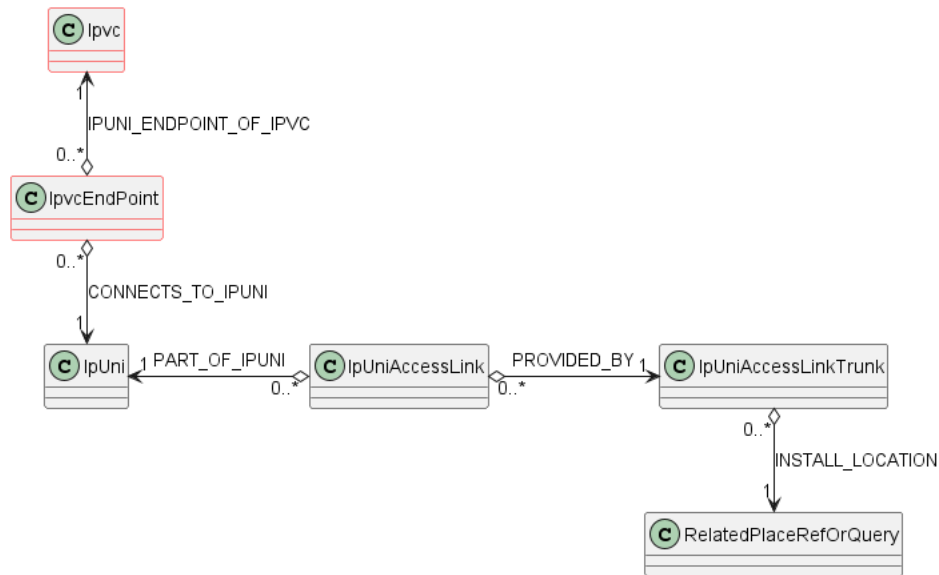


**Figure 8. A simplified view of IPVC End Point service specification data model**

Service specifications define several service-related and envelope-related requirements. For example:

- for an IPVC End Point service two mandatory relationships must be specified, one toward the IPVC (**IPUNI\_ENDPOINT\_OF\_IPVC**), and a second towards the IP UNI (**CONNECTS\_TO\_IPUNI**) for the **add** action.
- in the case of a **modify** action, service relationships must have the same value as in the **add** action. They must not be changed
- for an IP UNI Access Link Trunk service a place relationship (**INSTALL\_LOCATION**) must be specified
- in the case of a **modify** action, place relationships must have the same value as in the **add** action. They must not be changed

In case, some of these requirements are violated the SOF returns an error response to the BUS that indicates specific functional errors. These errors are listed in the response body (a list of **Error422** entries) for HTTP **422** response.



**Figure 9. Example use case configuration**

Figure 9 shows a setup of service configuration used by the example. The Advanced Internet Access is built from 5 services:

- IPVC
- IPVC End Point
- IP UNI
- IP UNI Access Link
- IP UNI Access Link Trunk

The example assumes a situation, where IP UNI, IP UNI Access Link, and IP UNI Access Link Trunk are already provisioned and are available in Service Inventory. They are marked with black lines. The Service Order includes requests to create 2 services: IPVC and IPVC End Point (marked with red lines). This means there are 2 Service Order Items with **action=add**. As mentioned earlier, there are 2 mandatory relations to be provided with IPVC End Point. In this case:

- **IPUNI\_ENDPOINT\_OF\_IPVC** is provided with the use of **serviceOrderItemRelationship** as pointing to the **Ipvc** being part of the same Service Order,
- **CONNECTS\_TO\_IPUNI** is provided with the use of **serviceRelationship** as pointing to an **IpUni** service that is already provisioned and available in Service Inventory.

## 5.5. Model Structural Validation

The structure of the payloads exchanged via Service API endpoints is defined using:

- OpenAPI version 3.0 for the service-agnostic part of the payload
- JsonSchema (draft 7) for the service-specific part of the payload

**[R6]** Implementations **MUST** use payloads that conform to these definitions.

**[R7]** The BUS and the SOF **MUST NOT** use any operation, entity or attribute that is not explicitly defined or allowed by this standard.

**[R8]** A service specification may define additional consistency rules and requirements that **MUST** be respected by implementations. These are defined for:

- required relation type, multiplicity to other items within the same or another Service Order request

- required relation type, multiplicity to entities in the SOF's service inventory
- related contact information roles that are to be defined at the Service Order Item level
- relations to places (locations) and their roles that are to be defined at the order item level

## 5.6. Security Considerations

There must be an authentication mechanism whereby a SOF can be assured who a BUS is and vice-versa. There must also be authorization mechanisms in place to control what a particular BUS or SOF is allowed to do and what information may be obtained. However, the definition of the exact security mechanism and configuration is outside the scope of this document. The LSO Security mechanisms are defined by *LSO API Security Profile* [[MEF 128.1](#)].

## 6. API Interactions and Flows

This section provides a detailed insight into the API functionality, use cases, and flows. It starts with Table 6 presenting a list and short description of all business use cases then presents the variants of end-to-end interaction flows, and in the following subchapters describes the API usage flow and examples for each of the use cases.

| Use Case # | Use Case Name                                      | Use Case Description                                                                                                                                                                                                                                                                                                                                       |
|------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | Create Service Order                               | A request initiated by the BUS to order a new service or service component(s). A Service Order must contain at least one Service Order Item (Use Case # 1-a, 1-b, or 1-c) as shown below. A Service Order may contain more than one Service Order Item and Service Order Items within a Service Order are not required to have relationships between them. |
| 1-a        | Service Order Item to Add Service                  | Service Order Item adds a new Service.                                                                                                                                                                                                                                                                                                                     |
| 1-b        | Service Order Item to Modify Existing Service      | Service Order Item modifies attributes of a specific active Service.                                                                                                                                                                                                                                                                                       |
| 1-c        | Service Order Item to Disconnect Existing Service  | Service Order Item disconnects an active Service.                                                                                                                                                                                                                                                                                                          |
| 2          | Retrieve List of Service Orders                    | A request initiated by the BUS to retrieve a list of Service Orders that match the provided filter criteria                                                                                                                                                                                                                                                |
| 3          | Retrieve Service Order by Service Order Identifier | A request initiated by the BUS to retrieve the details associated with a specific Service Order with the given Service Order Identifier.                                                                                                                                                                                                                   |
| 4          | Register for Notifications                         | The BUS requests to subscribe to notifications.                                                                                                                                                                                                                                                                                                            |
| 5          | Send Notification                                  | A notification initiated by the SOF to the BUS                                                                                                                                                                                                                                                                                                             |

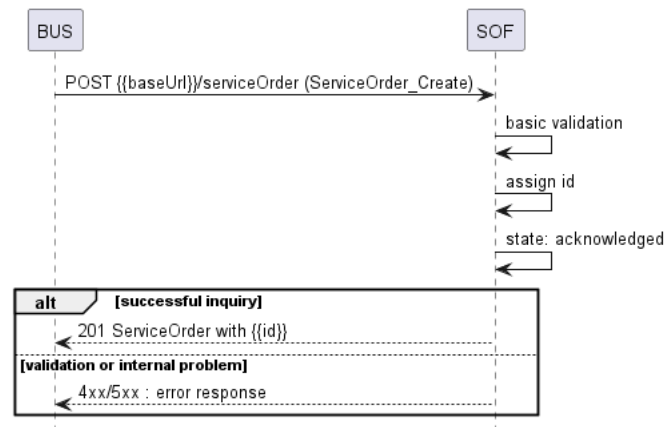
**Table 6. Use cases description**

### 6.1. Use case 1: Create Service Order

This is the initial step for Service Order processing.

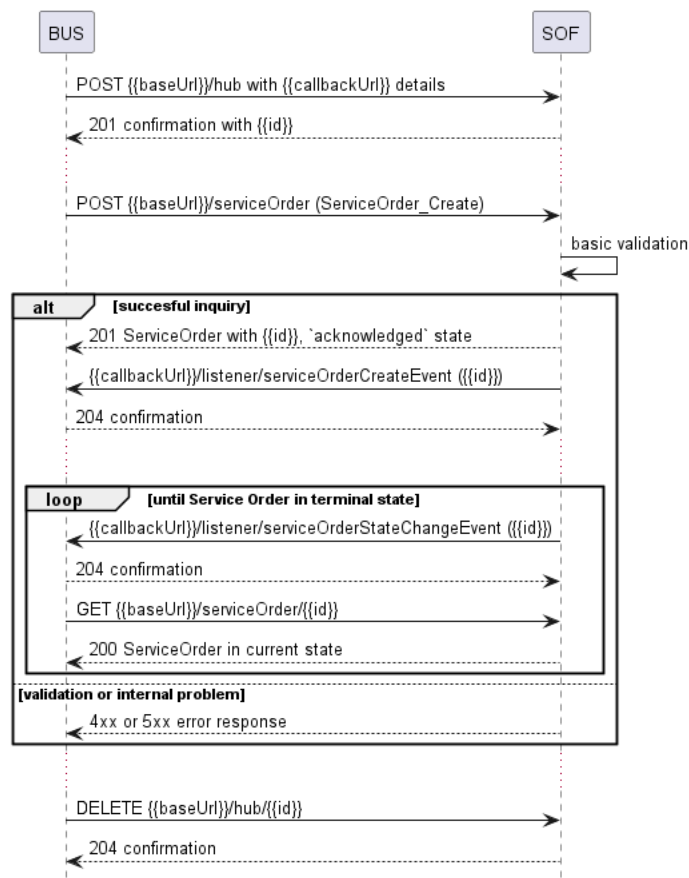
### 6.1.1. Interaction flow

The flow of this use case is very simple and is described in Figure 10.



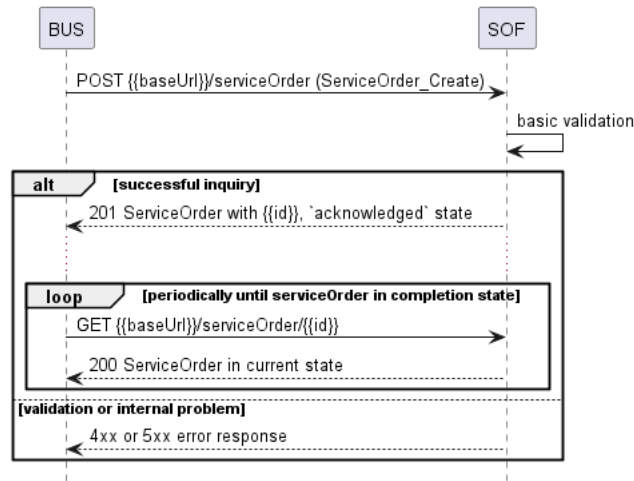
**Figure 10. Use Case 1 - Service Order create request flow**

The BUS sends a request with a **ServiceOrder\_Create** type in the body. The SOF performs request validation, assigns an **id**, and returns **ServiceOrder** type in the response body, with a **state** set to **acknowledged**. From this point, the Service Order is ready for further processing. The BUS can track the progress of the process either by subscribing for notifications or by periodically polling the **ServiceOrder**. The two patterns are presented in the following two diagrams.



**Figure 11. Service Order progress tracking - Notifications**





**Figure 12. Service Order progress tracking - Polling**

**Note:** The context of notifications is not a part of the considered use case itself. It is presented to show the big picture of end-to-end flow. This applies also to all further use case flow diagrams with notifications.

so to all further use case flow diagrams with notifications.

### 6.1.2. Create Service Order Request

Figure 13 presents the most important part of the data model used during the Create Service Order request (`POST /serviceOrder`) and response. The model of the request message - `ServiceOrder_Create` is a subset of the `ServiceOrder` model and contains only attributes that can (or must) be set by the BUS. The SOF then enriches the entity in the response with additional information.

**Note:** `ServiceOrder_Create` and `ServiceOrderItem_Create` are entities used by the BUS to make a request. `ServiceOrder` and `ServiceOrderItem` are entities used by the SOF to provide a response. The request entities have a subset of attributes of the response entities. Thus for visibility of these shared attributes `ServiceOrder_Common` and `ServiceOrderItem_Common` have been introduced. Though, these are not to be used directly in the exchange.

A `ServiceOrderItem_Create` defines details of the service(s) being subject of the ordering (in `ServiceValue` structure) and allows for the definition of additional information like related parties (`RelatedContactInformation`) or relations to other items (`ServiceOrderItemRelationship`, `ServiceOrderRelationship`).

`ServiceValue` allows for the introduction of service-specific properties as the Service Order payload. The extension mechanism is described in detail in [Section 5.3](#). `ServiceValue` may be also used to specify relations to places (using specializations of `RelatedPlaceOrValue`, as described in [Section 6.1.8](#). and/or to a service that exists in the SOF's inventory (using `ServiceRelationship`).

The full list of attributes is available in [Section 7](#) and in the API specification which is an integral part of this standard.



```

        "emailAddress": "BUS.ServiceOrderItemContact@client.mef.com",
        "name": "BUS Service Order Item Contact",
        "number": "+12-345-678-90",
        "role": "busServiceOrderItemContact"
    }
},
"serviceConfiguration": {
    "@type": "urn:mef:lso:spec:legato:ipvc:v0.0.1:all",
    "administrativeState": {
        "state": "UNLOCKED"
    },
    "operationalState": {
        "state": "ENABLED"
    },
    "ipvcIdentifier": "IPVC-0000-0001",
    "ipvcTopology": "CLOUD_ACCESS",
    "packetDelivery": "STANDARD_ROUTING",
    "maximumNumberOfIpv4Routes": 1,
    "maximumNumberOfIpv6Routes": 0,
    "dscpPreservation": "ENABLED",
    "serviceLevelSpecification": {}, <<skipped>>
    "maximumTransferUnit": 1522,
    "pathMtuDiscovery": "ENABLED",
    "fragmentation": "DISABLED",
    "cloud": {}, <<skipped>>
    "reservedPrefixes": {}, <<skipped>>
    "listOfClassOfServiceNames": ["low"]
}
},
{
    "id": "item-002",
    "action": "add",
    "serviceOrderItemRelationship": [
        {
            "orderItem": { << Relationship to IPVC in the same Service Order >>
                "itemId": "item-001"
            },
            "relationshipType": "IPUNI_ENDPOINT_OF_IPVC"
        }
    ],
    "service": {
        "description": "IPVC End Point",
        "externalId": "BUS_IPVC_END_POINT-0001",
        "serviceType": "Internet Access",
        "name": "IPVCEndpoint",
        "serviceRelationship": [
            { << Relationship to already configured IP UNI in Service Inventory >>
                "relationshipType": "CONNECTS_TO_IPUNI",
                "service": {
                    "id": "IP_UNI_0000-0001"
                }
            }
        ]
    },
    "relatedContactInformation": [
        {
            "emailAddress": "BUS.ServiceOrderItemContact@client.mef.com",
            "name": "BUS Service Order Item Contact",
            "number": "+12-345-678-90",
            "role": "busServiceOrderItemContact"
        }
    ],
    "serviceConfiguration": {
        "@type": "urn:mef:lso:spec:legato:ipvc-end-point:v0.0.1:all",
        "administrativeState": {
            "state": "UNLOCKED"
        },
        "operationalState": {
            "state": "ENABLED"
        },
        "identifier": "IPVC-EndPoint-0000-0001",
        "eiType": "UNI",
        "role": "ROOT",
        "prefixMapping": {},
        "maximumNumberOfIpv4Routes": 1,
        "maximumNumberOfIpv6Routes": 0,
        "ingressClassOfServiceMap": {}, <<skipped>>
        "egressClassOfServiceMap": {}, <<skipped>>
        "ingressBwpEnvelope": {}, <<skipped>>
        "egressBwpEnvelope": {} <<skipped>>
    }
}

```

```

    }
  }
}

```

[R9] The BUS's request **MUST** contain `requestedStartDate`, `requestedCompletionDate` and at least one `serviceOrderItem`.

[R10] The BUS's request **MUST** contain at least one `serviceOrderItem`.

[D1] The BUS and SOF **SHOULD** agree on using specific contact `roles`.

**Note:** During the onboarding the SOF may require to provide an additional contact `role`.

**Note:** It is up to SOF's discretion on how to react in case the BUS provides a contact `role` that is not agreed upon during the onboarding. Preferably the SOF should return an error with a message stating which `roles` are accepted. It may also be ignored

For each `serviceOrderItem`:

[R11] The BUS's Create Service Order request **MUST** contain:

- `id`
- `action`
- `service`

[R12] When adding a note, BUS **MUST** add a `note` only with `source=bus`.

### 6.1.3. Create Service Order Response

Entities use for providing a response to Create Service Order request are presented in Figure 13. The main types used for response are `ServiceOrder` and `ServiceOrderItem`, which add attributes set by SOF (like `id` or `state`) `ServiceOrder` is the root entity of a response. The response echoes back all attributes as provided by the BUS and contains the same number of `ServiceOrderItems` as in the request.

The following snippet presents the SOF's response.

#### Service Order Create Response

```

{
  "id": "00000000-3333-4444-5555-000000004567", << added by SOF >>
  "href": "{{baseUrl}}/serviceOrder/00000000-3333-4444-5555-000000004567", << added by SOF >>
  "state": "acknowledged", << added by SOF >>
  "orderDate": "2022-12-28T20:45:24.796Z", << added by SOF >>
  "expectedCompletionDate": "2023-01-25T20:00:00.000Z", << added by SOF >>
  "description": "Example Service Order",
  "externalId": "busOrder-101",
  "requestedCompletionDate": "2023-01-28T20:45:23.796Z",
  "requestedStartDate": "2023-01-02T00:00:00.000Z",
  "relatedContactInformation": [
    {
      "emailAddress": "john.example@client.mef.com",
      "name": "John Example",
      "number": "12-345-6789",
      "numberExtension": "1234",
      "organization": "Example Com.",
      "role": "serviceOrderContact"
    },
    { << added by SOF >>
      "emailAddress": "ella.sof@seller.mef.com",
      "name": "Ella SOF",
      "number": "98-765-4321",
    }
  ]
}

```

```

    "organization": "SOF Co.",
    "role": "sofContact"
  }
],
"note": [
  {
    "id": "note-001",
    "author": "John Example",
    "date": "2022-12-28T20:45:23.796Z",
    "source": "bus",
    "text": "This is an example text"
  },
  { << added by SOF >>
    "id": "note-002",
    "author": "Ella SOF",
    "date": "2022-12-28T20:45:24.796Z",
    "source": "sof",
    "text": "This is an example response text"
  }
],
"serviceOrderItem": [
  {
    "id": "item-001",
    "action": "add",
    "state": "acknowledged", << added by SOF >>
    "service": {
      "id": "00000000-5555-6666-7777-000000008888", << added by SOF >>
      "href": "{{baseUrl}}/service/00000000-5555-6666-7777-000000008888", << added by SOF >>
      "state": "feasibilityChecked",
      "description": "IP Virtual Connection",
      "externalId": "BUS_IPVC-0001",
      "serviceType": "Internet Access",
      "name": "IPVC"
      ...
      << skipped, as provided by BUS >>
    }
  },
  {
    "id": "item-002",
    "action": "add",
    "state": "acknowledged", << added by SOF >>
    "serviceOrderItemRelationship": [
      {
        "orderItem": {
          "itemId": "item-001",
          "serviceOrderHref": "string",
          "serviceOrderId": "string"
        },
        "relationshipType": "IPUNI_ENDPOINT_OF_IPVC"
      }
    ],
    "service": {
      "id": "00000000-5555-6666-7777-000000009999", << added by SOF >>
      "href": "{{baseUrl}}/service/00000000-5555-6666-7777-000000009999", << added by SOF >>
      "state": "feasibilityChecked",
      "description": "IPVC End Point",
      "externalId": "BUS_IPVC_END_POINT-0001",
      "serviceType": "Internet Access",
      "name": "IPVCEndpoint",
      "serviceRelationship": [
        {
          "relationshipType": "CONNECTS_TO_IPUNI",
          "service": {
            "id": "IP_UNI_0000-0001"
          }
        }
      ]
      ...
      << skipped, as provided by BUS >>
    }
  }
]
]

```

Attributes that are set by the SOF in the response are marked with the **<< added by SOF >>** tag. The response to the create request does not contain all possible attributes. Some of them are valid only in the future lifecycle of the **ServiceOrder** (e.g. **completionDate**, **startDate**).

[R13] The SOF's response **MUST** include all and unchanged attributes' values as provided by BUS in the request.

The SOF might append related contact information or notes if required, but cannot modify items set by the BUS.

[R14] The SOF **MUST** specify the following attributes in a response:

- `id`
- `state`
- `orderDate`

[R15] The `id` **MUST** remain the same value for the life of the Service Order.

[R16] When adding a note, SOF **MUST** add a `note` only with `source=sof`.

[R17] Notes **MUST NOT** be modified or deleted once entered.

For each `serviceOrderItem`:

[R18] The response **MUST** have the `state` attribute set.

[R19] If the Service Order Item `state` in the SOF's response is not `completed`, the response **MUST NOT** contain the `completionDate`.

#### 6.1.4. Use Case 1a: Service Order Item to Add Service

When requesting a new service installation (`action` equal to `add`) the BUS needs to provide all of its configuration information. The example for `add` action is already provided in the snippets above.

The following requirements apply when `serviceOrderItem.action` is `add`:

[R20] The BUS **MUST** provide:

- `service.state`
- `service.serviceConfiguration`

[R21] If there is a relationship with a Service Order Item within the same Service Order, the `serviceOrderItemRelationship.itemId` **MUST** be specified.

[R22] If there is a relationship with a Service Order Item within the same Service Order, the `serviceOrderItemRelationship.itemId` and `serviceOrderItemRelationship.serviceOrderId` **MUST NOT** be specified.

[R23] If there is a relationship with a Service Order Item of another Service Order, the `serviceOrderItemRelationship.itemId` and `serviceOrderItemRelationship.serviceOrderId` **MUST** be specified.

[R24] The BUS **MUST NOT** specify the `serviceOrderItem.service.id` in the request. It is the SOF who assigns this id.

*Note:* The `service.id` might not be assigned yet at the moment the SOF provides a response for the Create Service Order Request.

#### 6.1.5. Use case 1b: Service Order Item to Modify Existing Service

The following example shows a request for an order for an existing IPVC End Point Service modification (**action** equal to **modify**). In particular, a change to **maximumNumberOfIpv4Routes** is introduced.

The IPVC End Point service exists in SOF's inventory and is identified as **00000000-5555-6666-7777-000000009999**, as provided in SOF response presented in [Chapter 6.1.3](#).

The following requirements apply to **serviceOrderItem** when **action** is **modify**:

[R25] The modify request **MUST** specify a reference (provide **service.id**) to an existing service that is a subject of this order and provide the desired **service.serviceConfiguration**.

[R26] The modify request **MUST** provide:

- **service.id** - a reference to an existing service that is a subject of this order
- **service.state**
- **service.serviceConfiguration**

[R27] The BUS **MUST** send the full **serviceOrderItem** including a full **serviceOrderItem.service**, containing all attributes that are expected to be set once the **serviceOrderItem** is successfully fulfilled, even if some of them remain unchanged.

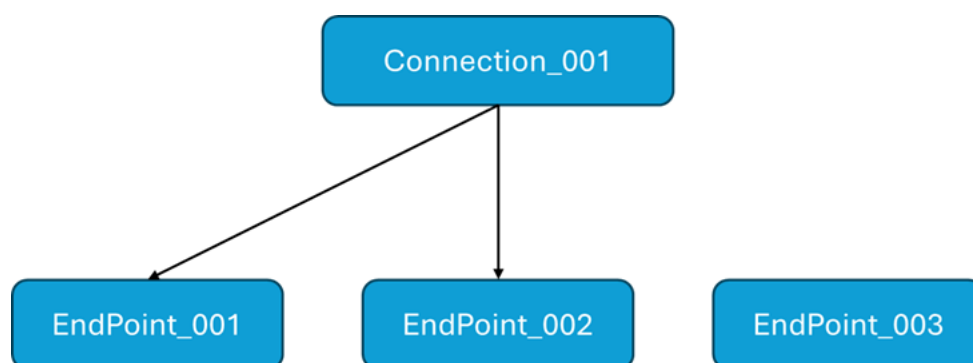
[R28] The BUS request **MUST** comply to the rules of modification defined by Service specification (incl. whether **serviceRelationship** and/or **place** can be modified).

[O3] The SOF **MAY** introduce additional (to ones defined by Service specification) business constraints on which attributes can be modified.

[R29] If the modification request violates any of modification restriction rules, the SOF **MUST** return an appropriate error response (422) to the BUS.

[R30] The modification request **MUST** contain the full "to-be" list of Service Relationships ( that eventually will be stored in the Service Inventory as **service.serviceRelationship**). This is expressed by a combination of **serviceOrderItem.service.serviceRelationship** and **serviceOrderItem.serviceOrderItemRelationship**.

Following examples provide a guide on how the requests should look like in use cases of adding or removing relations to existing or added Services.



**Figure 14. Service relationship modification - starting point.**

Figure 14 presents a starting point for each of the examples. It is a simplified view on the Service Inventory with 4 Services present (imaginary model). One "Connection" and three "EndPoints". The **relationshipType** between these Services types is called **TERMINATES\_ON**. Arrows represent existing Service relationships. The assumption is that the modification of the relationships is allowed and the number of relationships is not restricted. The example payloads show only a subset of attributes relevant to use cases. Even mandatory parameter are hidden.

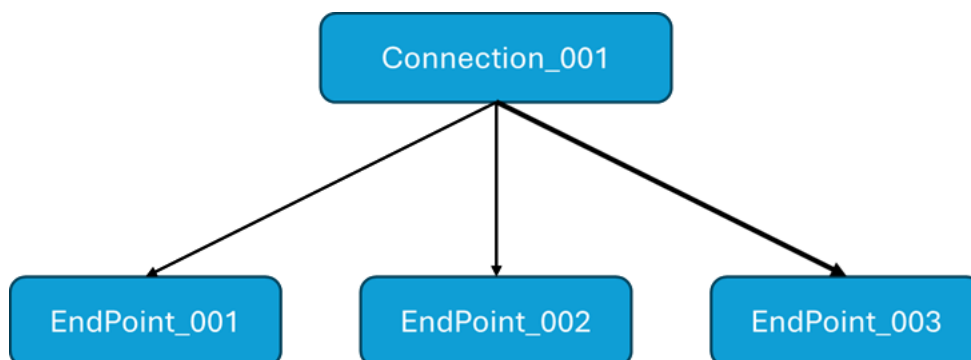
**Note:** Each example builds on Figure 14 as starting point.

### Example 1:

Add service relationship to a Service already existing in the Service Inventory.

When adding a relationship to an existing Service - `serviceOrderItem.service.serviceRelationship` is used.

```
{
  //other Service Order attributes
  "serviceOrderItem": [
    {
      "id": "item-1",
      "action": "modify",
      "service": {
        "id": "Connection_001",
        "serviceRelationship": [
          {
            "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
            "service": {
              "id": "EndPoint_001"
            }
          },
          {
            "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
            "service": {
              "id": "EndPoint_002"
            }
          },
          {
            "relationshipType": "TERMINATES_ON", // To be added
            "service": {
              "id": "EndPoint_003"
            }
          }
        ]
      }
    }
  ],
  //other service attributes
},
"serviceOrderItemRelationship": []
//other service order item attributes
}
]
```



**Figure 15. Service relationship modification - Example 1 outcome.**

### Example 2:

Add service relationship to a Service created in the same Service Order as modification request.

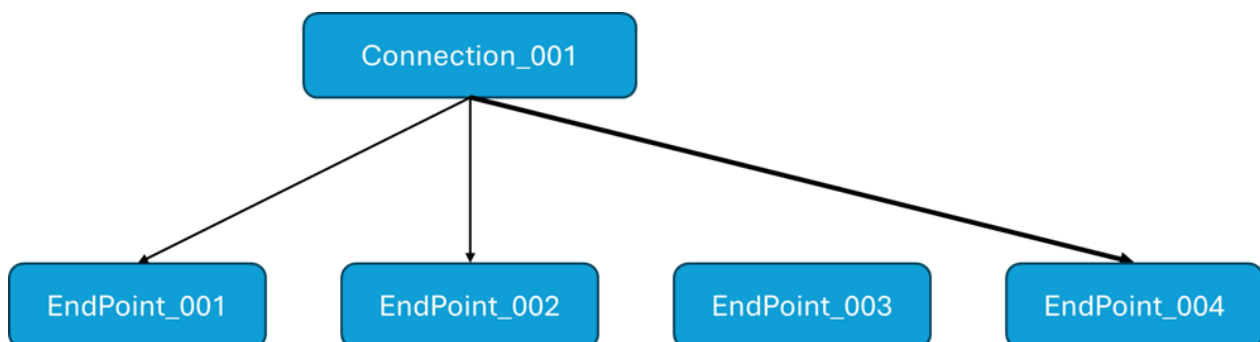
When adding a relationship to newly created Service - `serviceOrderItem.serviceOrderItemRelationship` is used.



```

{
  //other Service Order attributes
  "serviceOrderItem": [
    {
      "id": "item-1",
      "action": "modify",
      "service": {
        "id": "Connection_001",
        "serviceRelationship": [
          {
            "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
            "service": {
              "id": "EndPoint_001"
            }
          },
          {
            "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
            "service": {
              "id": "EndPoint_002"
            }
          }
        ]
      },
      //other service attributes
    },
    "serviceOrderItemRelationship": [
      // To be added - relation to item creating new Service
      {
        "relationshipType": "TERMINATES_ON",
        "orderItem": {
          "itemId": "item-2"
        }
      }
    ]
  ],
  //other service order item attributes
},
{
  "id": "item-2",
  "action": "add",
  "service": {
    "state": "active",
    "serviceRelationship": []
    //other service attributes
  },
  "serviceOrderItemRelationship": [],
  //other service order item attributes
}
]
}

```



**Figure 16. Service relationship modification - Example 2 outcome.**

### **Example 3:**

Delete the existing relationship.

```

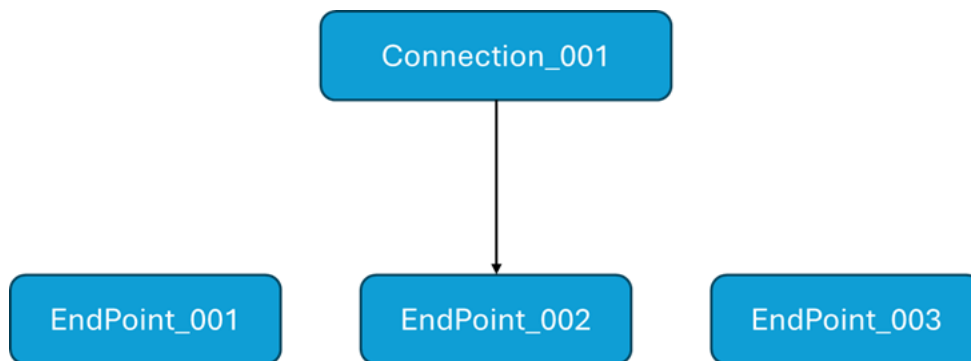
{
  //other Service Order attributes
  "serviceOrderItem": [
    {
      "id": "1",

```

```

    "action": "modify",
    "service": {
      "id": "Connection_001",
      "serviceRelationship": [
        // Relation to "Endpoint_001" missing
        {
          "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
          "service": {
            "id": "Endpoint_002"
          }
        }
      ]
    },
    //other service attributes
  },
  "serviceOrderItemRelationship": []
  //other service order item attributes
}
]
}

```



**Figure 17. Service relationship modification - Example 3 outcome.**

#### Example 4:

This example shows a combination of all previous examples in one request:

- Add service relationship to a Service already existing in the Service Inventory.
- Add service relationship to a Service created in the same Service Order as modification request.
- Delete the existing relationship.

```

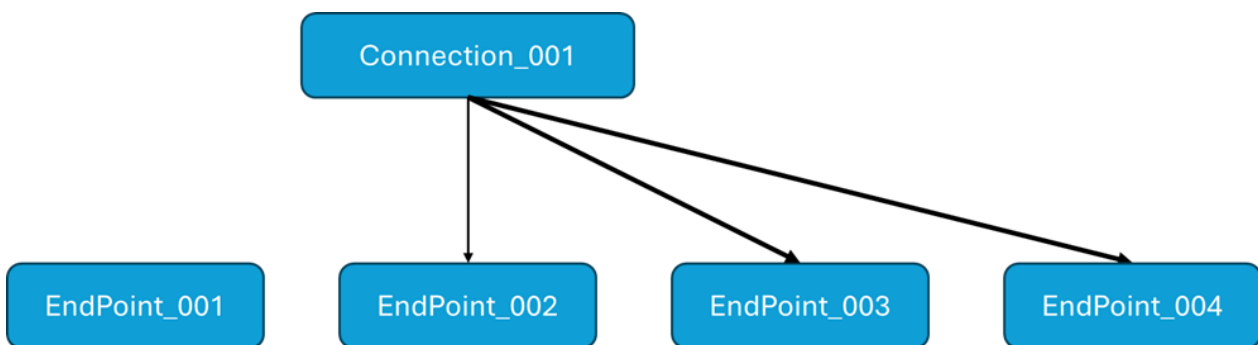
{
  //other Service Order attributes
  "serviceOrderItem": [
    {
      "id": "item-1",
      "action": "modify",
      "service": {
        "id": "Connection_001",
        "serviceRelationship": [
          // Relation to "Endpoint_001" missing
          {
            "relationshipType": "TERMINATES_ON", // Existing in Service Inventory
            "service": {
              "id": "Endpoint_002"
            }
          },
          {
            "relationshipType": "TERMINATES_ON", // To be added
            "service": {
              "id": "Endpoint_003"
            }
          }
        ]
      }
    }
  ],
  //other service attributes
},
"serviceOrderItemRelationship": [
  // To be added - relation to item creating new Service

```

```

    {
      "relationshipType": "TERMINATES_ON",
      "orderItem": {
        "itemId": "item-2"
      }
    }
  ]
  //other service order item attributes
},
{
  "id": "item-2",
  "action": "add",
  "service": {
    "state": "active",
    "serviceRelationship": []
    //other service attributes
  },
  "serviceOrderItemRelationship": []
  //other service order item attributes
}
]
}

```



**Figure 18. Service relationship modification - Example 4 outcome.**

Please also note, that in the **add** case, a reference to the IPVC service used the **serviceOrderItemRelationship** pointing to another **serviceOrderItem** in the same Service Order Request. This is because the IPVC did not exist at that moment and was also a part of the order. In the case of ordering the update of an existing IPVC End Point, the IPVC is also existing and it must be referenced with the use of **serviceRelationship**. This example below assumes that the IPVC service is available in SOF's Inventory with the **id** equals "**00000000-5555-6666-7777-000000008888**" (as provided in SOF response presented in [chapter 6.1.3](#)).

### Service Order Item to Modify Existing Service

```

{
  "description": "Example Service Order to Modify IPVC End Point service",
  "externalId": "busOrder-102",
  "requestedCompletionDate": "2023-02-03T20:45:23.796Z",
  "requestedStartDate": "2023-02-02T00:00:00.000Z",
  "relatedContactInformation": [
    {
      "emailAddress": "john.example@client.mef.com",
      "name": "John Example",
      "number": "12-345-6789",
      "numberExtension": "1234",
      "organization": "Example Co.",
      "role": "serviceOrderContact"
    }
  ],
  "serviceOrderItem": [
    {
      "id": "item-001",
      "action": "modify",
      "service": {
        "id": "00000000-5555-6666-7777-000000009999", << id to point to service instance >>
        "description": "IPVC End Point",
        "externalId": "BUS_IPVC_END_POINT-0001",

```

```

"serviceType": "Internet Access",
"name": "IPVCEndpoint",
"state": "active",
"serviceRelationship": [
  { << relation to IP UNI - not changed >>
    "relationshipType": "CONNECTS_TO_IPUNI",
    "service": {
      "id": "IP_UNI_0000-0001"
    }
  },
  { << relation to IPVC - not changed, but provided with serviceRelationship instead of
serviceOrderItemRelationship >>
    "relationshipType": "IPUNI_ENDPOINT_OF_IPVC",
    "service": {
      "id": "00000000-5555-6666-7777-000000008888"
    }
  }
],
"serviceConfiguration": {
  "@type": "urn:mef:lso:spec:legato:ipvc-end-point:v0.0.1:all",
  "administrativeState": {
    "state": "UNLOCKED"
  },
  "operationalState": {
    "state": "ENABLED"
  },
  "identifien": "IPVC-EndPoint-0000-0001",
  "eiType": "UNI",
  "role": "ROOT",
  "prefixMapping": {},
  "maximumNumberOfIpv4Routes": 2, << modified value >>
  "maximumNumberOfIpv6Routes": 0,
  "ingressClassOfServiceMap": {},
  "egressClassOfServiceMap": {},
  "ingressBwpEnvelope": {},
  "egressBwpEnvelope": {}
}
}
}
]
}

```

### 6.1.6. Use case 1c: Service Order Item to Disconnect Existing Service

The example below represents a single Service Order request for disconnect (**action=delete**) of an existing IPVC End Point service.

The effect of the **action=delete** request is only a disconnect of the Service ("logical deletion"). The Service transitions to **terminated** state. In effect this action is equal to using a request with **action=modify** and **state=terminated**.

#### Service Order to Disconnect Existing Service

```

{
  "description": "Example Service Order to Disconnect IPVC End Point service",
  "externalId": "busOrder-103",
  "requestedCompletionDate": "2023-03-03T20:45:23.796Z",
  "requestedStartDate": "2023-03-02T00:00:00.000Z",
  "serviceOrderItem": [
    {
      "id": "item-001",
      "action": "delete",
      "service": {
        "id": "00000000-5555-6666-7777-000000009999" << id to point to service instance >>
      }
    }
  ]
}

```

The following requirements apply to **serviceOrderItem** when **action** is **delete**:

[R31] `service.id` **MUST** be provided.

[R32] The BUS **MUST NOT** provide any `service` attributes other than `service.id`.

### 6.1.7. Service Order and Service Order Items State Machine

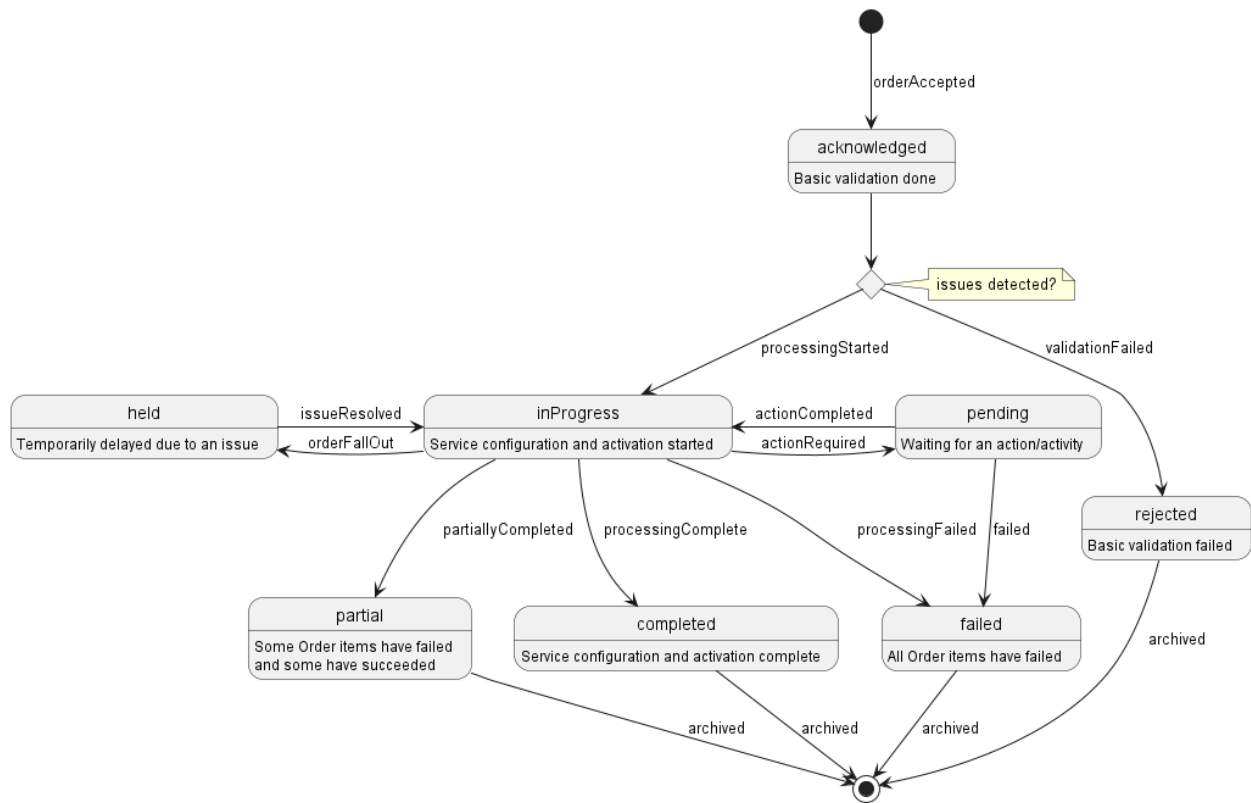


Figure 19. Service Order State Machine

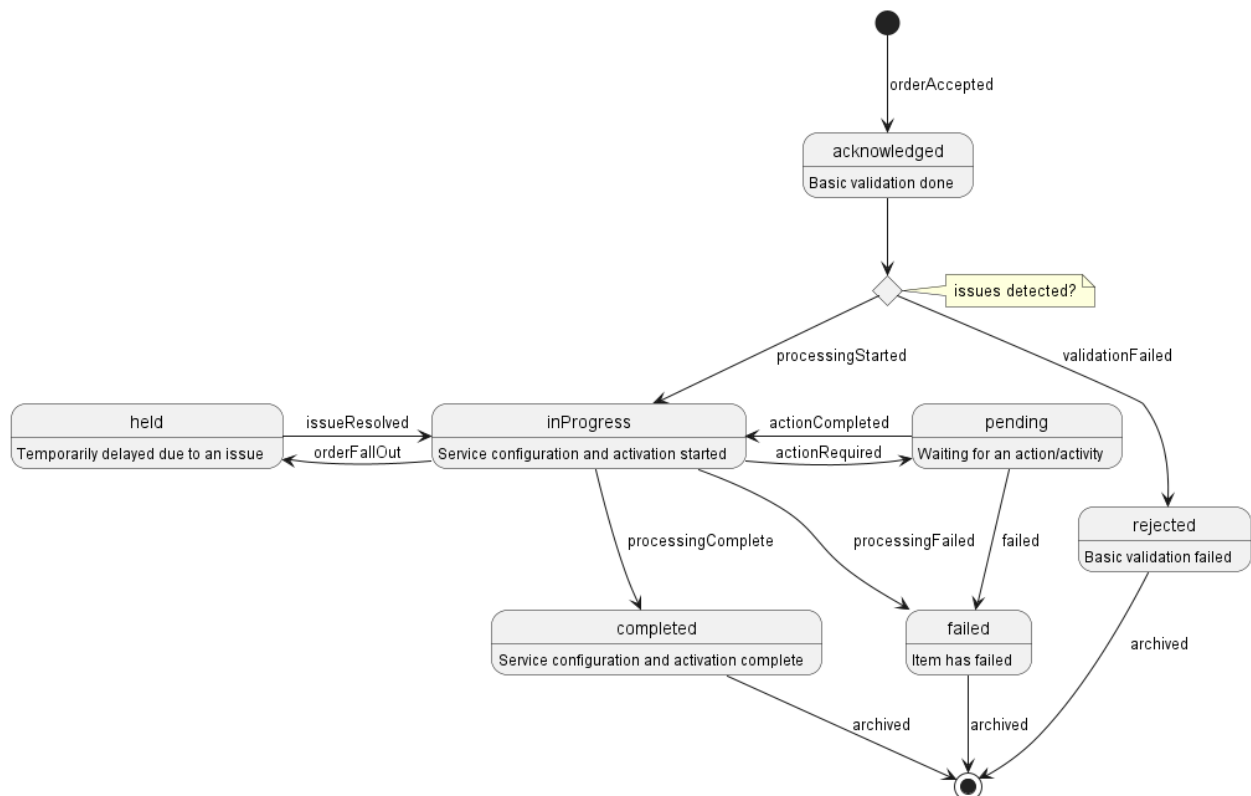


Figure 20. Service Order Item State Machine

Service Order and Service Order Item share almost the same list of possible states and states' transitions (Service Order Item does not have the **partial** state available). They are presented in Figures 14 and 15.

After receiving the request, the SOF performs basic checks of the message. If any problem is found an Error response is provided. If the validation passes a response is provided with **ServiceOrder** and all **ServiceOrderItems** in the **acknowledged** state. Before moving the order to the **inProgress** state, the BUS performs all the remaining business and time-consuming validations. At this point, an Error response cannot be provided anymore so the order moves to a **rejected** state if some issues are found. The **serviceOrderItem.terminationError** acts as a placeholder to provide a detailed description of what caused the problem.

Tables 7 and 8 present the Service Order and Service Order Item states' descriptions.

| State        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acknowledged | A <b>ServiceOrder</b> request has been received and has passed message and basic validations and a <i>Success Response</i> has been sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| rejected     | This state indicates that:<br>- Invalid information is provided through the <b>ServiceOrder</b> / <b>ServiceOrderItem</b> request<br>- The request fails to meet validation rules for <b>Service</b> delivery (processing)<br>If one <b>ServiceOrderItem</b> is <b>rejected</b> , then the entire <b>ServiceOrder</b> request is <b>rejected</b> and a <i>Error Response</i> is sent.                                                                                                                                                                                                               |
| inProgress   | This state indicates that all <b>ServiceOrderItems</b> have successfully passed the validations checks and the scheduled <b>Service</b> delivery/processing has started. The <b>ServiceOrder</b> will be in <b>inProgress</b> state if <i>at least one ServiceOrderItem</i> is in <b>inProgress</b> state                                                                                                                                                                                                                                                                                           |
| pending      | This state indicates that a <b>ServiceOrderItem</b> is currently in a waiting stage for an action/activity to be completed before the order-processing can progress further (this may happen also via non-API channel).<br>A <b>pending</b> state can lead into automatic <b>failed</b> of an <b>ServiceOrderItem</b> , if no action is taken within the agreed timeframe.<br>The <b>ServiceOrder</b> will be in <b>pending</b> state if <i>at least one ServiceOrderItem</i> is in <b>pending</b> state                                                                                            |
| held         | This state indicates that a <b>ServiceOrderItem</b> cannot be progressed due to an issue. The <b>Service</b> delivery (processing) has been temporarily delayed to resolve an infrastructure shortfall to facilitate the supply of order. Upon resolution of the issue, the <b>ServiceOrderItem</b> will continue to progress.<br>A <b>held</b> state can lead into automatic <b>failed</b> of a <b>ServiceOrderItem</b> if no action is taken within the agreed timeframe.<br>The <b>ServiceOrder</b> will be in <b>held</b> state if at least one <b>ServiceOrderItem</b> is in <b>held</b> state |
| failed       | This state indicates that <b>Service</b> delivery (processing) associated with a <b>ServiceOrderItem</b> has failed. This indicates an irrecoverable error as opposed to <b>held</b> or <b>pending</b> issues.<br>The <b>ServiceOrder</b> will be in <b>failed</b> state if at <i>ALL ServiceOrderItems</i> are in <b>failed</b> state                                                                                                                                                                                                                                                              |
| completed    | This state indicates that <b>Service</b> delivery (processing) associated with a <b>ServiceOrderItem</b> has completed.<br>The <b>ServiceOrder</b> will be in <b>completed</b> state if at <i>ALL ServiceOrderItems</i> are in <b>completed</b> state                                                                                                                                                                                                                                                                                                                                               |

| State   | Description                                                                                                                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| partial | This state indicates that some <i>ServiceOrderItem</i> are in <i>completed</i> state while others are in <i>failed</i> states, so the entire <i>ServiceOrder</i> is in a <i>partial</i> state. Not applicable to <i>ServiceOrderItem</i> . |

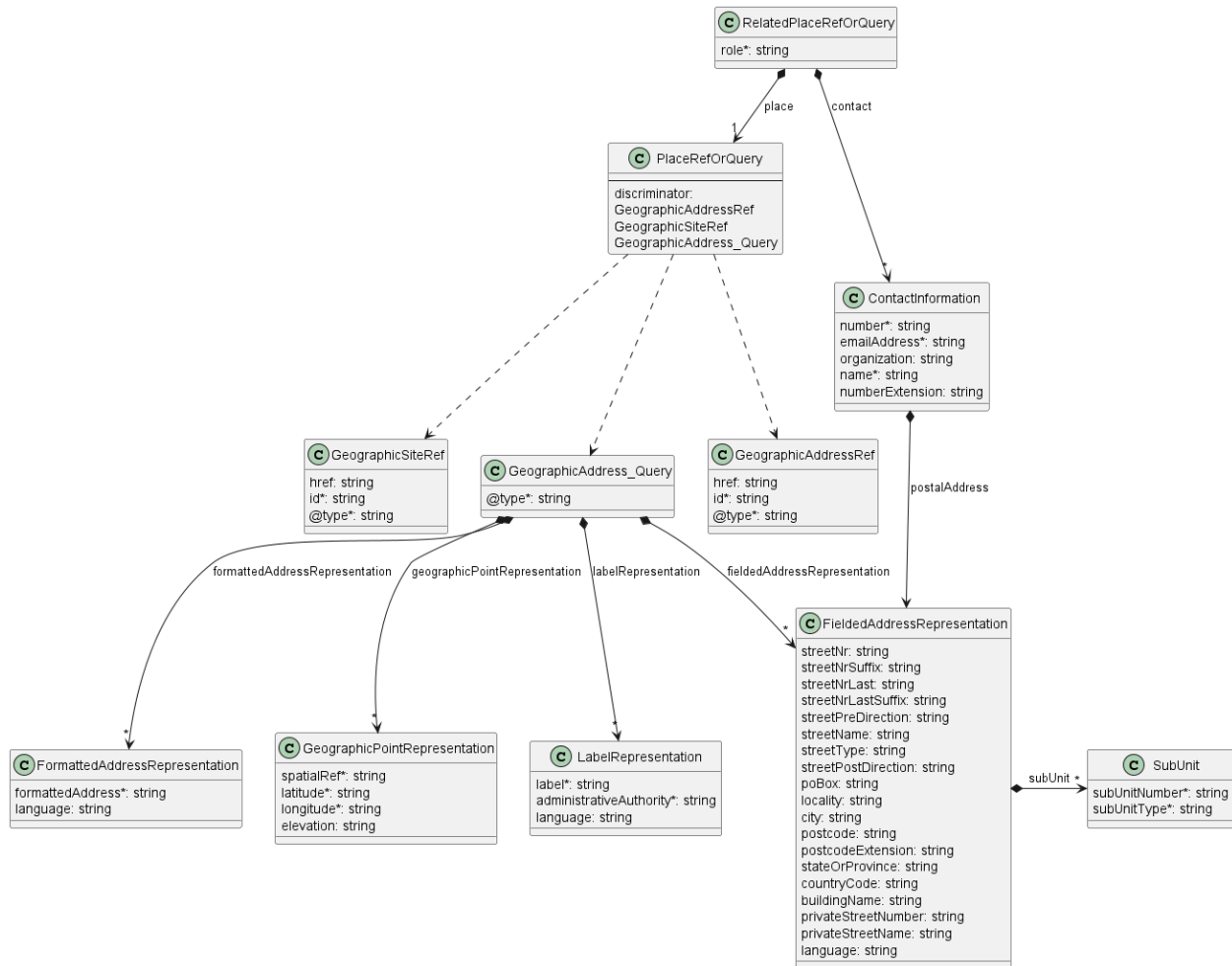
**Table 7. Service Order states**

| State        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| acknowledged | A <i>ServiceOrder</i> request has been received and has passed message and basic validations and a <i>Success Response</i> has been sent.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| rejected     | This state indicates that:<br>- Invalid information is provided through the <i>ServiceOrderItem</i> request<br>- The request fails to meet validation rules for <i>Service</i> delivery (processing)<br>If one <i>ServiceOrderItem</i> is rejected, then the entire <i>ServiceOrder</i> request is rejected and a <i>Error Response</i> is sent.                                                                                                                                                                                                                                                   |
| inProgress   | This state indicates that <i>ServiceOrderItem</i> have successfully passed the validations checks and the scheduled <i>Service</i> delivery/processing has started.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| pending      | This state indicates that a <i>ServiceOrderItem</i> is currently in a waiting stage for an action/activity to be completed before the order-processing can progress further (this may happen also via non-API channel).<br>A <i>pending</i> state can lead into automatic <i>failed</i> of an <i>ServiceOrderItem</i> , if no action is taken within the agreed timeframe.<br>The <i>ServiceOrder</i> will be in <i>pending</i> state if <i>at least one ServiceOrderItem</i> is in <i>pending</i> state                                                                                           |
| held         | This state indicates that a <i>ServiceOrderItem</i> cannot be progressed due to an issue. The <i>Service</i> delivery (processing) has been temporarily delayed to resolve an infrastructure shortfall to facilitate supply of order. Upon resolution of the issue, the <i>ServiceOrderItem</i> will continue to progress.<br>A <i>held</i> state can lead into automatic <i>failed</i> of an <i>ServiceOrderItem</i> , if no action is taken within the agreed timeframe.<br>The <i>ServiceOrder</i> will be in <i>held</i> state if at least one <i>ServiceOrderItem</i> is in <i>held</i> state |
| failed       | This state indicates that <i>Service</i> delivery (processing) associated with a <i>ServiceOrderItem</i> has failed. This indicates an irrecoverable error as opposed to <i>held</i> or <i>pending</i> issues.<br>The <i>ServiceOrder</i> will be in <i>failed</i> state if at <i>ALL ServiceOrderItems</i> are in <i>failed</i> state                                                                                                                                                                                                                                                             |
| completed    | This state indicates that <i>Service</i> delivery (processing) associated with a <i>ServiceOrderItem</i> has completed.<br>The <i>ServiceOrder</i> will be in <i>completed</i> state if at <i>ALL ServiceOrderItems</i> are in <i>completed</i> state                                                                                                                                                                                                                                                                                                                                              |

**Table 8. Service Order Item states**

### 6.1.8. Providing the place information

When required by service specification, the Service must point to the place where the Service is provided. This is done with the use of the *place* attribute of type *RelatedPlaceRefOrQuery*, which is presented in Figure 21.



**Figure 21. Data model - referring to a place**

The **role** defines the function that the place plays for a given Service. The name of the role to be provided is strictly defined by the service specification. Usually, it is **INSTALL\_LOCATION**.

**contact** provides additional information about the person to contact to get access to this place in case such access is required to complete the evaluation of this Quote Item.

**place** is where the actual place is pointed. The attribute is of type **PlaceRefOrQuery** which is an abstract class that can be of one of three types: **GeographicAddressRef**, **GeographicSiteRef**, or **GeographicAddress\_Query**. The first two are simple identifiers to reference a **GeographicAddress** or **GeographicSite** respectively. The BUS usually first validates the **GeographicAddress** and gets its identifier from the SOF and then optionally retrieves **GeographicSite** information for that address. In the unlikely case that the SOF does not provide the Address Validation API and the BUS is not able to obtain the address identifier in any other way, the **GeographicAddressQuery** type might be used. It contains lists of Geographic Address Representations to provide the address information by value. There are four types of Geographic Address Representations:

- **FieldedAddressRepresentation**
- **FormattedAddressRepresentation**
- **LabelRepresentation**
- **GeographicPointRepresentation**

One or more of these representations may be used to describe a single place.

The **GeographicAddress** model together with its above-mentioned representations and respective requirements are defined by [Mplify 121.1](#) (chapter 5.3). That standard is the owner of those definitions. This API specification contains a model of **GeographicAddress** but does not define it.



Any further changes of these types will update the API specification, but will not be reflected in this document.

The mandatory `@type` attribute of `GeographicSiteRef`, `GeographicAddressRef` and `GeographicAddress_Query` is used as a discriminator to unambiguously identify the intended type when using in the context of the `oneOf` section of `PlaceRefOrQuery` type.

## 6.2. Use Case 2: Retrieve List of Service Orders

The BUS can retrieve a list of `ServiceOrders` by using a `GET /serviceOrder` operation with desired filtering criteria.

[O4] The BUS's request **MAY** contain none or more of the following attributes:

- `state`
- `orderDate.gt`
- `orderDate.lt`
- `completionDate.gt`
- `completionDate.lt`
- `expectedCompletionDate.gt`
- `expectedCompletionDate.lt`
- `startDate.gt`
- `startDate.lt`

A response to retrieve a list of results can be paginated. The BUS can specify following query attributes related to pagination:

- `limit` - number of expected list items
- `offset` - offset of the first element in the result list

The filtering and pagination attributes must be specified in URI query format [RFC3986](#). The SOF returns a list of elements that comply with the requested `limit`. If the requested `limit` is higher than the supported list size the smaller list result is returned. In that case, the size of the result is returned in the header attribute `X-Result-Count`. The SOF can indicate that there are additional results available using:

- `X-Total-Count` header attribute with the total number of available results
- `X-Pagination-Throttled` header set to `true`

[D1] The Seller **SHOULD** support the pagination mechanism.

[CR1]<[D1] Seller **MUST** use either `X-Total-Count` or `X-Pagination-Throttled` to indicate that the page was truncated and additional results are available.

```
https://serverRoot/mefApi/legato/serviceOrderingManagement/v6/serviceOrder?state=completed&limit=10&offset=0
```

The example above shows a BUS's request to get all `ServiceOrders` that are in the `completed` state. Additionally, the BUS asks only for a first (`offset=0`) pack of 10 results (`limit=10`) to be returned. The correct response (HTTP code `200`) in the response body contains a list of `ServiceOrder` objects matching the criteria.

[R33] In case no items matching the criteria are found, the SOF **MUST** return a valid response with an empty list.

## 6.3. Use Case 3: Retrieve Service Order by Service Order Identifier

The BUS can get detailed information about the Service Order from the SOF by using a **GET /serviceOrder/{id}** operation. The payload returned in the response includes all attributes the BUS has provided while sending a Service Order create request. The attributes provided by the SOF depend on the status of the **ServiceOrder** and may require some time to be set.

Both Get List and Get by Identifier operations return the same **ServiceOrder** representation, so a response to a get by id for a **ServiceOrder** with **id=00000000-3333-4444-5555-000000004567** would return exactly the same response as presented in [section 6.1.3](#).

**[R34]** In case **id** does not allow finding a **ServiceOrder** in SOF's system, an error response **Error404 MUST** be returned.

**[R35]** Once the service identifier (**serviceOrder.serviceOrderItem.service.id**) is assigned, it **MUST** be provided in the SOF's response.

## 6.4. Use case 4: Register for Notifications

The SOF communicates with the BUS with Notifications provided that:

- BUS supports a notification mechanism
- BUS has registered to receive notifications from the SOF

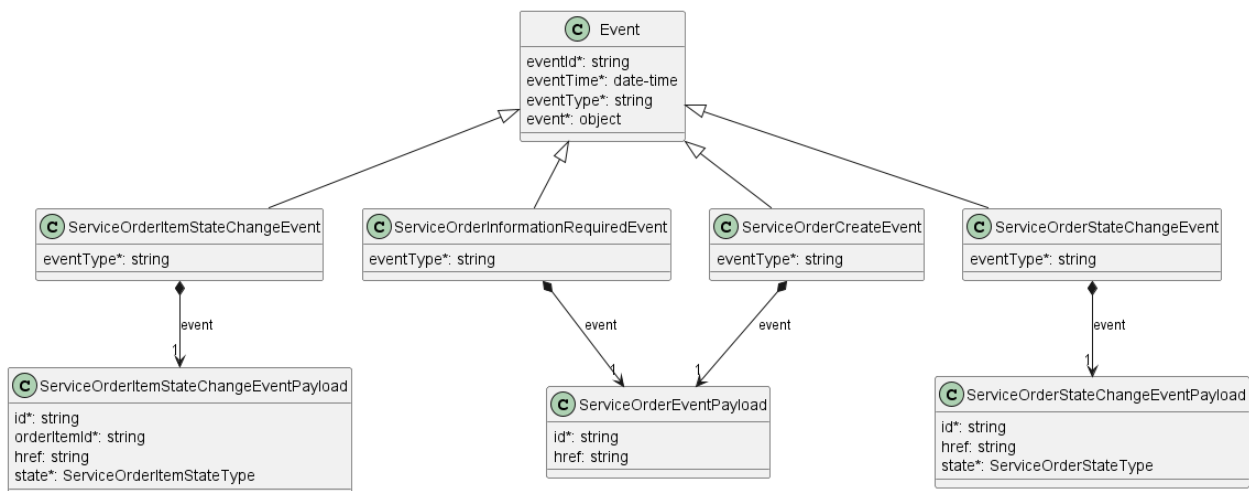
**[O5]** BUS **MAY** register for Notifications.

Supporting Notification is mandatory for SOF.

To register for notifications the BUS uses the **registerListener** operation from the API: **POST /hub**. The request contains only 2 attributes:

- **callback** - mandatory, to provide the callback address the events will be notified to,
- **query** - optional, to provide the required types of event.

Figure 22 shows all entities involved in the Notification use cases.



**Figure 22. Service Ordering Notification Data Model**

By using a simple request:

```
{
  "callback": "https://client.mef.com/listenerEndpoint"
}
```

The BUS subscribes for notification of all types of events. Those are:

- `serviceOrderCreateEvent`
- `serviceOrderStateChangeEvent`
- `serviceOrderItemStateChangeEvent`
- `serviceOrderInformationRequiredEvent`

If the BUS wishes to receive only notifications of a certain type, a `query` must be added:

```
{
  "callback": "https://client.mef.com/listenerEndpoint",
  "query": "eventType=serviceOrderStateChangeEvent"
}
```

If the BUS wishes to subscribe to 2 different types of events, there are 2 possible syntax variants [TMF630]:

```
eventType=serviceOrderStateChangeEvent,serviceOrderItemStateChangeEvent
```

or

```
eventType=serviceOrderStateChangeEvent&eventType=serviceOrderItemStateChangeEvent
```

The `query` formatting complies with RFC3986 [RFC3986](#). According to it, every attribute defined in the Event model (from notification API) can be used in the `query`. However, this standard requires only `eventType` attribute to be supported.

[R36] `eventType` is the only attribute that the SOF **MUST** support in the query.

The SOF responds to the subscription request by adding the `id` of the subscription to the message that must be further used for unsubscribing.

```
{
  "id": "00000000-0000-0000-0000-000000000678",
  "callback": "https://client.mef.com/listenerEndpoint",
  "query": "eventType=serviceOrderStateChangeEvent"
}
```

Example of a final address that the Notifications will be sent to (for `serviceOrderStateChangeEvent`):

- `https://client.mef.com/listenerEndpoint/mefApi/legato/serviceOrderingNotification/v6/listener/serviceOrderStateChangeEvent`

## 6.5. Use case 5: Send Notification

Notifications are used to asynchronously inform the BUS about the respective objects and attributes changes.

For sake of readability, all previous flow diagrams presented only cases of using only the `serviceOrderStateChangeEvent`. Figure 23 presents the an end-to-end sequence of

communication in Use Case 1 - Create Service Order with BUS's subscription to both **serviceOrderStateChangeEvent** and **serviceOrderItemStateChangeEvent** event types.

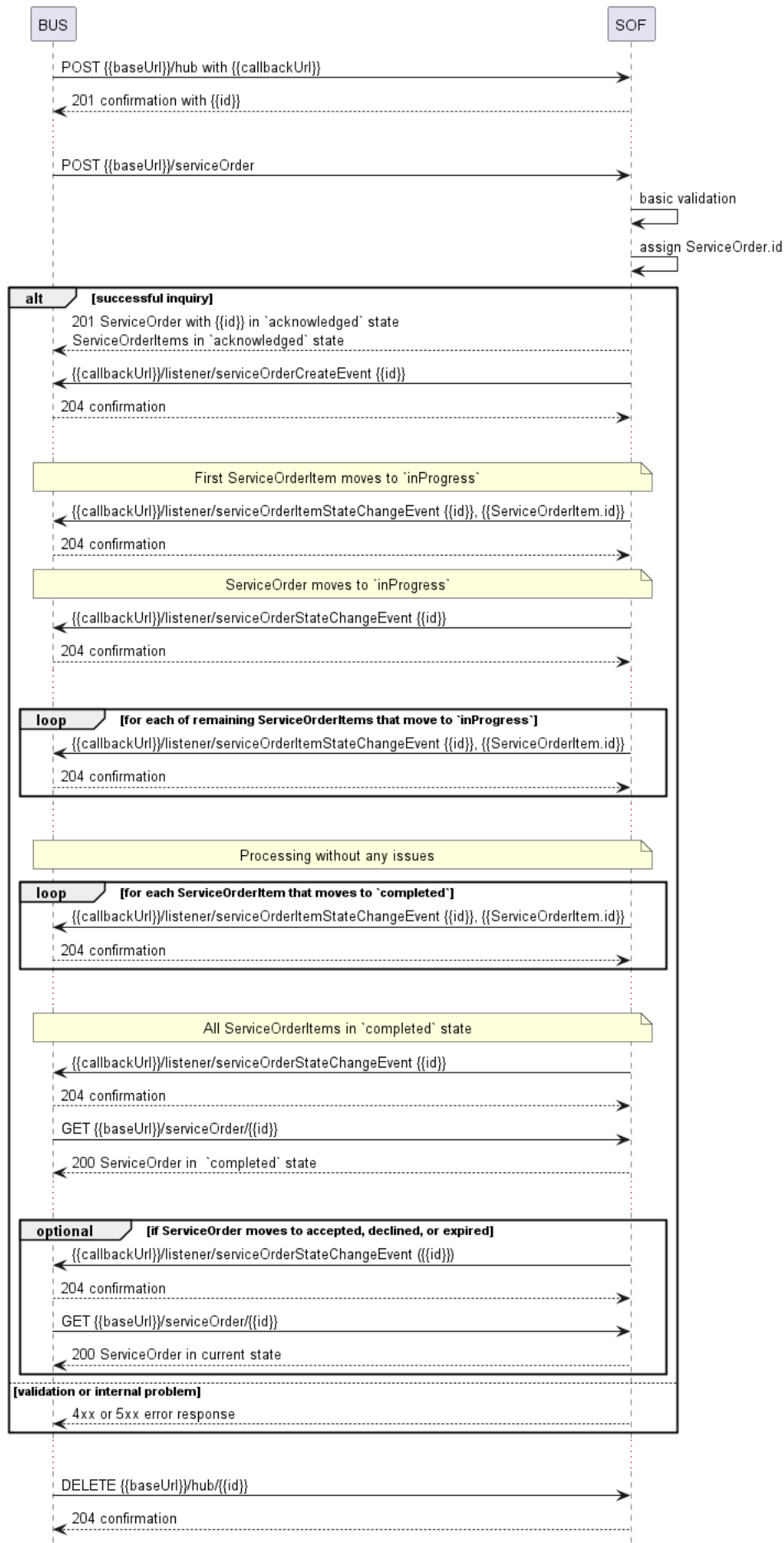


Figure 23. Use Case 1 - Create Service Order with all Notifications

After a successful Notification subscription, the BUS sends a Service Order create request. The SOF responds with Service Order and all items in **acknowledged** state. Creation of Service Order is notified with a **serviceOrderCreateEvent**. When the first Service Order Item moves to **inProgress**, a **serviceOrderItemStateChangeEvent** is sent. Immediately the Service Order also changes its state to **inProgress** and the **serviceOrderStateChangeEvent** is sent. Then the rest (if any) of the Service Order Items are processed. When particular items are done processing they reach the **completed** state. Once all are successfully done, the Service Order also changes state to **completed**. The BUS will likely now ask for the Service Order details.

**Note:** The state change notification are sent only when the state attribute actually changes its value. There are no status change notifications sent upon Service Order or Service Order Item creation.

[R37] The SOF **MUST NOT** send Notifications to BUS that have not registered for them.

[R38] The SOF **MUST** send Notifications to BUS that have registered for them.

Following snippets present examples of **serviceOrderStateChangeEvent** and **serviceOrderItemStateChangeEvent**:

```
{
  "eventId": "event-001",
  "eventType": "serviceOrderStateChangeEvent",
  "eventTime": "2022-12-28T20:45:24.796Z",
  "event": {
    "id": "00000000-3333-4444-5555-000000004567",
    "state": "inProgress"
  }
}
```

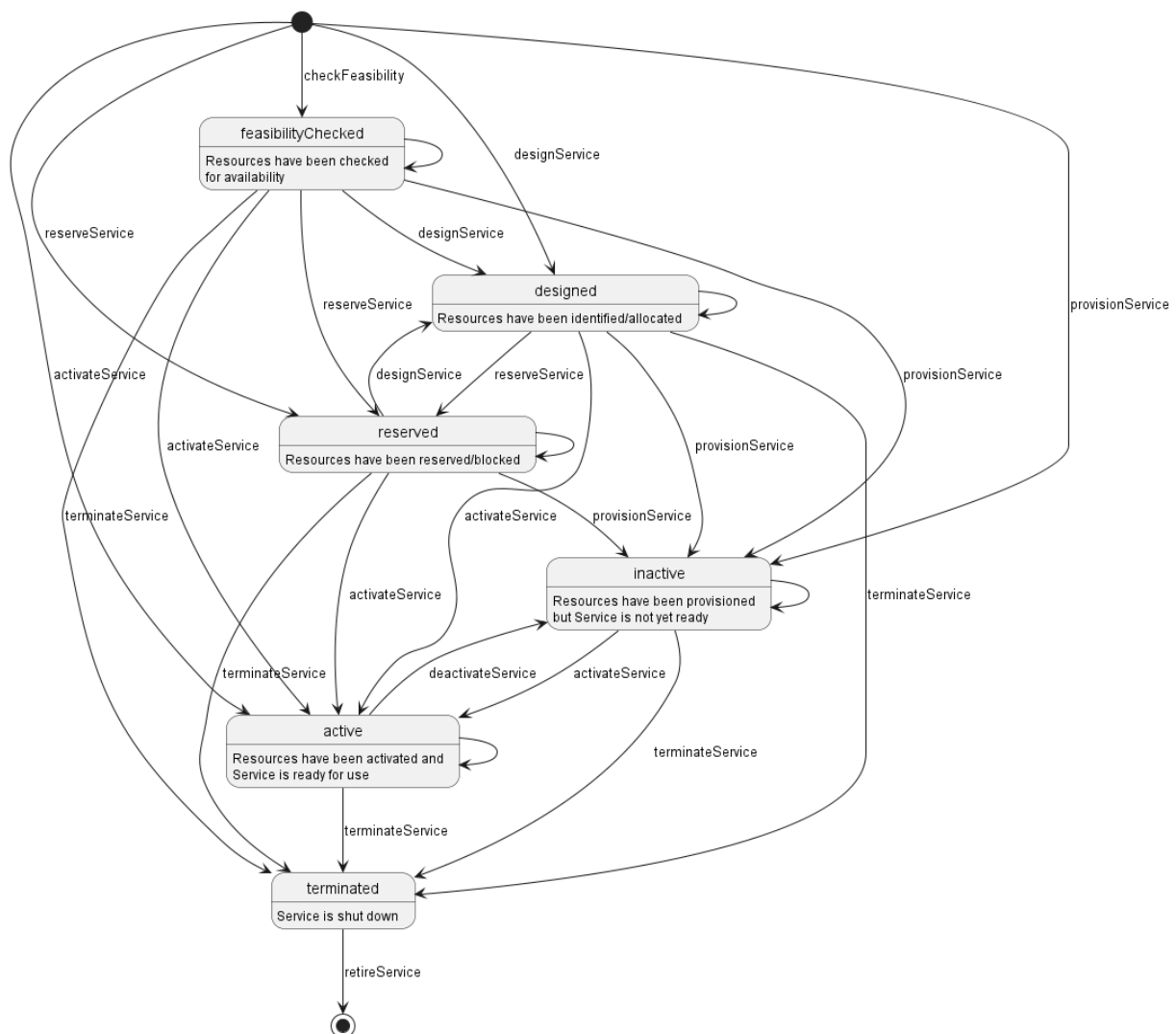
[R39] An event triggered by the Service Order Item (**serviceOrderItemStateChangeEvent**) **MUST** additionally contain the relative **orderItemId**.

```
{
  "eventId": "event-002",
  "eventType": "serviceOrderItemStateChangeEvent",
  "eventTime": "2023-01-15T20:45:24.796Z",
  "event": {
    "id": "00000000-3333-4444-5555-000000004567",
    "orderItemId": "item-001",
    "state": "inProgress"
  }
}
```

To stop receiving events, the BUS has to use the **unregisterListener** operation from the **DELETE /hub/{id}** endpoint. The **id** is the identifier received from the SOF during the listener registration.

## 6.6. Service Lifecycle

Above chapters focus on the requirements and the lifecycle of `ServiceOrder` and `ServiceOrderItem`. It is also very important to understand the lifecycle of the `Service` itself and how to manage it with the Service Ordering.



### Figure 24. Service Lifecycle

Figure 24 depicts the Service available states and their transitions.

The service is created by a request `action=add` and a desired `state`. All but `terminated` can be the initial state.

BUS can order Service state transition by placing a `ServiceOrderItem` with `action=modify` and providing the desired `service.state` attribute. State transitions form sort of use cases that can be performed on a Service. They are gathered in Table 9 together with requirements on the Service state they are applicable for. A `modify` request does not have to change the Service state. This is indicated as a loop arrow on Figure 24.

A request with `action=delete` acts as a `terminateService` use case.

| Use case         | action | state              | pre-condition |
|------------------|--------|--------------------|---------------|
| checkFeasibility | add    | feasibilityChecked | N/A           |
| designService    | add    | designed           | N/A           |

| Use case         | action | state      | pre-condition                                                    |
|------------------|--------|------------|------------------------------------------------------------------|
|                  | modify | designed   | feasibilityChecked<br>reserved                                   |
| reserveService   | add    | reserved   | N/A                                                              |
|                  | modify | reserved   | feasibilityChecked<br>designed                                   |
| provisionService | add    | inactive   | N/A                                                              |
|                  | modify | inactive   | feasibilityChecked<br>designed<br>reserved                       |
| activateService  | add    | active     | N/A                                                              |
|                  | modify | active     | feasibilityChecked<br>designed<br>reserved<br>inactive           |
| deactivate       | modify | inactive   | active                                                           |
| terminateService | modify | terminated | feasibilityChecked<br>designed<br>reserved<br>inactive<br>active |
|                  | delete | N/A        | feasibilityChecked<br>designed<br>reserved<br>inactive<br>active |

**Table 9. Service Life cycle Use Cases**

It is up to the Seller's discretion on what is the retention period of Service being in the **terminated** state.

Table 10 summarizes the states and their descriptions:

| State              | Description                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| feasibilityChecked | Initial check whether the necessary resources are available and sufficient for the installation of a given service.   |
| designed           | The Service is designed. The resources are identified and/or allocated, but not reserved.                             |
| reserved           | All required resources for given service are reserved and ready.                                                      |
| inactive           | The service is deactivated and is no longer available.                                                                |
| active             | The service is fully available and active                                                                             |
| terminated         | The service is 'logically deleted'. All associated resources are freed and made available for service to other users. |

**Table 10. Service states**

## 7. API Details

### 7.1. API patterns

#### 7.1.1. Indicating errors

Erroneous situations are indicated by appropriate HTTP responses. An error response is indicated by HTTP status 4xx (for client errors) or 5xx (for server errors) and appropriate response payload. The Service Order API uses the error responses as depicted and described below.

Implementations can use HTTP error codes not specified in this standard in compliance with rules defined in RFC 7231 [RFC7231]. In such a case, the error message body structure might be aligned with the **Error**.

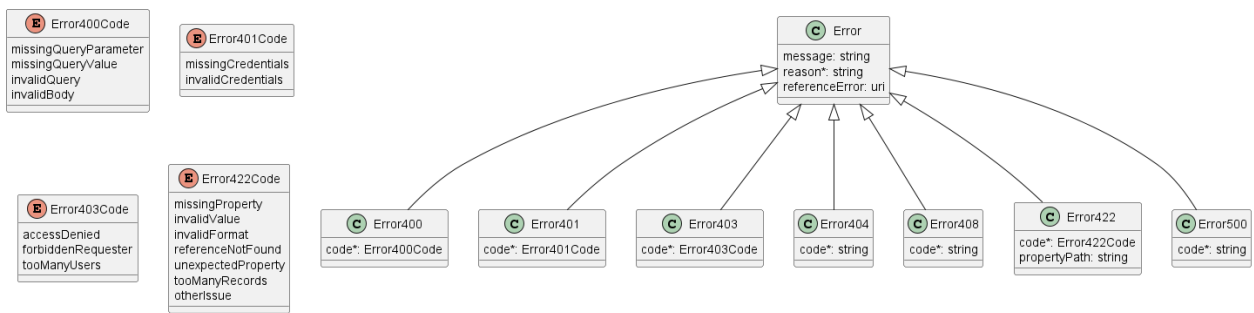


Figure 25. Data model types to represent an erroneous response

##### 7.1.1.1. Type Error

**Description:** Standard Class used to describe API response error Not intended to be used directly. The **code** in the HTTP header is used as a discriminator for the type of error returned in runtime.

| Name           | Type   | Description                                                                                                      |
|----------------|--------|------------------------------------------------------------------------------------------------------------------|
| message        | string | Text that provides mode details and corrective actions related to the error. This can be shown to a client user. |
| reason*        | string | Text that explains the reason for the error. This can be shown to a client user.                                 |
| referenceError | uri    | URL pointing to documentation describing the error                                                               |

##### 7.1.1.2. Type Error400

**Description:** Bad Request. (<https://tools.ietf.org/html/rfc7231#section-6.5.1>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                                                                                                                                                                         |
|-------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error400Code</a> | One of the following error codes: - missingQueryParam: The URI is missing a required query-string parameter - missingQueryValue: The URI is missing a required query-string parameter value - invalidQuery: The query section of the URI is invalid. - invalidBody: The request has an invalid body |



### 7.1.1.3. **enum** Error400Code

**Description:** One of the following error codes:

- missingQueryParameter: The URI is missing a required query-string parameter
- missingQueryValue: The URI is missing a required query-string parameter value
- invalidQuery: The query section of the URI is invalid.
- invalidBody: The request has an invalid body

### 7.1.1.4. **Type** Error401

**Description:** Unauthorized. (<https://tools.ietf.org/html/rfc7235#section-3.1>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                        |
|-------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error401Code</a> | One of the following error codes: - missingCredentials: No credentials provided. - invalidCredentials: Provided credentials are invalid or expired |

### 7.1.1.5. **enum** Error401Code

**Description:** One of the following error codes:

- missingCredentials: No credentials provided.
- invalidCredentials: Provided credentials are invalid or expired

### 7.1.1.6. **Type** Error403

**Description:** Forbidden. This code indicates that the server understood the request but refuses to authorize it. (<https://tools.ietf.org/html/rfc7231#section-6.5.3>)

Inherits from:

- [Error](#)

| Name  | Type                         | Description                                                                                                                                                                                                                                |
|-------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error403Code</a> | This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes: - accessDenied: Access denied - forbiddenRequester: Forbidden requester - tooManyUsers: Too many users |

### 7.1.1.7. **enum** Error403Code

**Description:** This code indicates that the server understood the request but refuses to authorize it because of one of the following error codes:

- accessDenied: Access denied
- forbiddenRequester: Forbidden requester
- tooManyUsers: Too many users

### 7.1.1.8. Type Error404

**Description:** Resource for the requested path not found.  
(<https://tools.ietf.org/html/rfc7231#section-6.5.4>)

Inherits from:

- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|       |        |                                                                                                  |
|-------|--------|--------------------------------------------------------------------------------------------------|
| code* | string | The following error code: - notFound: A current representation for the target resource not found |
|-------|--------|--------------------------------------------------------------------------------------------------|

### 7.1.1.9. Type Error422

The response for HTTP status **422** is a list of elements that are structured using the **Error422** data type. Each list item describes a business validation problem. This type introduces the **propertyPath** attribute which points to the erroneous property of the request, so that the BUS may fix it easier. It is highly recommended that this property should be used, yet remains optional because it might be hard to implement.

**Description:** Unprocessable entity due to a business validation problem.  
(<https://tools.ietf.org/html/rfc4918#section-11.2>)

Inherits from:

- [Error](#)

| Name | Type | Description |
|------|------|-------------|
|------|------|-------------|

|       |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | <a href="#">Error422Code</a> | One of the following error codes: - missingProperty: The property that was expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the target system - unexpectedProperty: Additional, not expected property has been provided - tooManyRecords: the number of records to be provided in the response exceeds the threshold. - otherIssue: Other problem was identified (detailed information provided in a reason) |
|-------|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|              |        |                                                                                                                                                                                                                                                                                                    |
|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| propertyPath | string | A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer ( <a href="https://tools.ietf.org/html/rfc6901">https://tools.ietf.org/html/rfc6901</a> ). |
|--------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 7.1.1.10. **enum** Error422Code

**Description:** One of the following error codes:

- missingProperty: The property that was expected is not present in the payload
- invalidValue: The property has an incorrect value
- invalidFormat: The property value does not comply with the expected value format

- `referenceNotFound`: The object referenced by the property cannot be identified in the target system
- `unexpectedProperty`: Additional, not expected property has been provided
- `tooManyRecords`: the number of records to be provided in the response exceeds the threshold.
- `otherIssue`: Other problem was identified (detailed information provided in a reason)

### 7.1.1.11. Type Error500

**Description:** Internal Server Error. (<https://tools.ietf.org/html/rfc7231#section-6.6.1>)

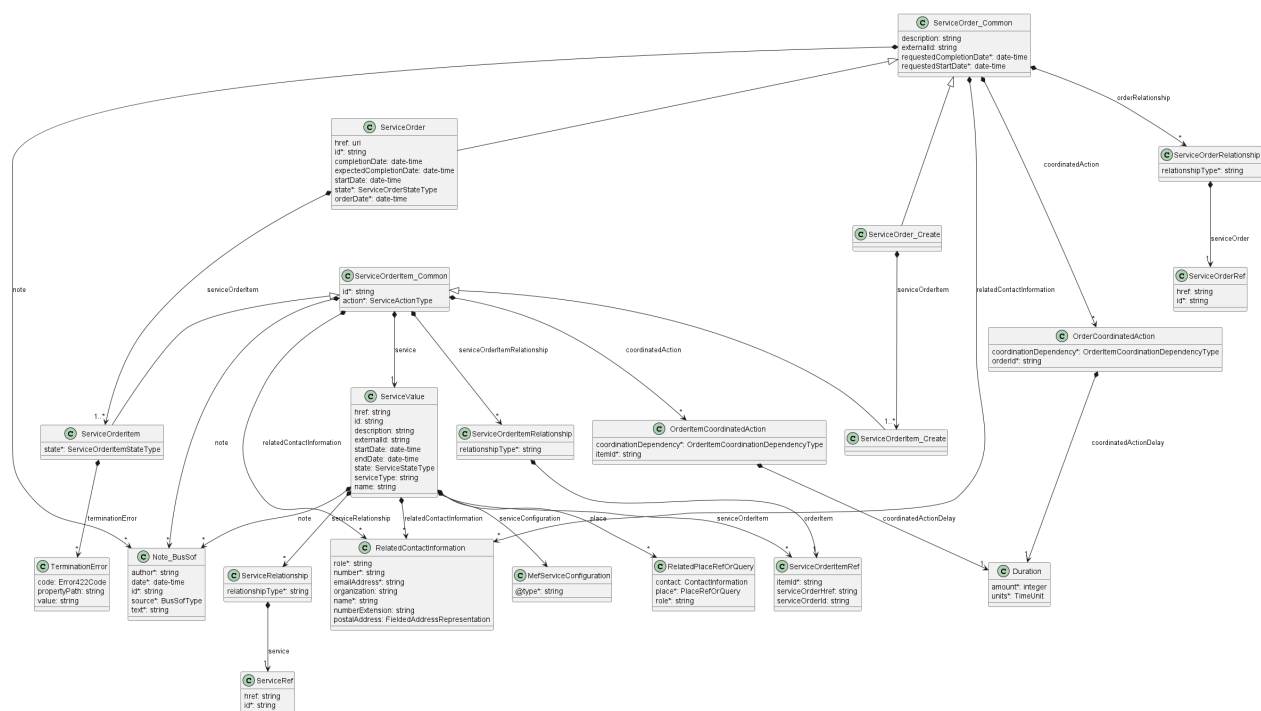
Inherits from:

- Error

| Name  | Type   | Description                                                                                                                                                                    |
|-------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code* | string | The following error code: - <code>internalError</code> : Internal server error - the server encountered an unexpected condition that prevented it from fulfilling the request. |

## 7.2. Management API Data model

Figure 26 presents the Service Order Management data model. The data types are discussed later in this section.



### Figure 26. Service Order Management Data Model

### 7.2.1. ServiceOrder

### 7.2.1.1 Type ServiceOrder\_Common

**Description:** A Service Order is used to request operations on a Service instance. A Service Order groups one or more one Service Order Items - one per specific action on a Service instance. The Action associated with the Service Order Item describes the operation (add, modify, delete) to be applied on the specified Service instance. The Service Order Item and its associated Action

can operate on both existing (modify, delete) as well as future (add) Service instance. The Service Order is triggered from the Business Application (BA) system in charge of the Service Order management to the Service Orchestration Function (SOF) system that will orchestrate the Service fulfillment.

This type defines all attributes common to objects used in request and response.

| Name                      | Type                                         | M/O | Description                                                                                                                                                              |
|---------------------------|----------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| coordinatedAction         | <a href="#">OrderCoordinatedAction</a> []    | O   | The interval after the completion of one or more related Service Order Items that this Service Order Item can be started or completed                                    |
| description               | string                                       | O   | A free-text description of the service order                                                                                                                             |
| externalId                | string                                       | O   | ID given by the consumer to facilitate searches                                                                                                                          |
| note                      | <a href="#">Note_BusSof</a> []               | O   | Extra-information about the order; e.g. useful to add extra delivery information that could be useful for a human process                                                |
| orderRelationship         | <a href="#">ServiceOrderRelationship</a> []  | O   | A list of service orders related to this order                                                                                                                           |
| relatedContactInformation | <a href="#">RelatedContactInformation</a> [] | O   | Contact information of an individual or organization playing a role for this Service Order. For providing Notification Contact, 'role=notificationContact' MUST be used. |
| requestedCompletionDate   | date-time<br><i>format = date-time</i>       | M   | Requested delivery date from the requestors perspective                                                                                                                  |
| requestedStartDate        | date-time<br><i>format = date-time</i>       | M   | Order start date wished by the requestor                                                                                                                                 |

### 7.2.1.2. Type ServiceOrder\_Create

**Description:** A Service Order is used to request operations on a Service instance. A Service Order groups one or more one Service Order Items - one per specific action on a Service instance. The Action associated with the Service Order Item describes the operation (add, modify, delete) to be applied on the specified Service instance. The Service Order Item and its associated Action can operate on both existing (modify, delete) as well as future (add) Service instance. The Service Order is triggered from the Business Application (BA) system in charge of the Service Order management to the Service Orchestration Function (SOF) system that will orchestrate the Service fulfillment. This type extends [ServiceOrder\\_Common](#) and adds attributes specific to the request response.

Inherits from:

- [ServiceOrder\\_Common](#)

| Name             | Type                                                              | M/O | Description                                                 |
|------------------|-------------------------------------------------------------------|-----|-------------------------------------------------------------|
| serviceOrderItem | <a href="#">ServiceOrderItem_Create</a> []<br><i>minItems = 1</i> | M   | A list of service order items to be processed by this order |

### 7.2.1.3. Type ServiceOrder

**Description:** A Service Order is used to request operations on a Service instance. A Service Order groups one or more one Service Order Items - one per specific action on a Service instance. The Action associated with the Service Order Item describes the operation (add, modify, delete) to be applied on the specified Service instance. The Service Order Item and its associated Action can operate on both existing (modify, delete) as well as future (add) Service instance. The Service Order is triggered from the Business Application (BA) system in charge of the Service Order management to the Service Orchestration Function (SOF) system that will orchestrate the Service fulfillment.

Inherits from:

- [ServiceOrder\\_Common](#)

| Name                   | Type                                                       | M/O | Description                                                                                             |
|------------------------|------------------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------|
| href                   | uri                                                        | O   | Hyperlink reference                                                                                     |
| id                     | string                                                     | M   | unique identifier                                                                                       |
| completionDate         | date-time<br><i>format = date-time</i>                     | O   | Effective delivery date amended by the provider                                                         |
| expectedCompletionDate | date-time<br><i>format = date-time</i>                     | O   | Expected delivery date amended by the provider                                                          |
| serviceOrderItem       | <a href="#">ServiceOrderItem</a> []<br><i>minItems = 1</i> | M   | A list of service order items to be processed by this order                                             |
| startDate              | date-time<br><i>format = date-time</i>                     | O   | Date when the order was started for processing                                                          |
| state                  | <a href="#">ServiceOrderStateType</a>                      | M   | The state of the Service Order                                                                          |
| orderDate              | date-time<br><i>format = date-time</i>                     | M   | Date when the Service Order was created in the SOF's system and a Service Order Identifier was assigned |

### 7.2.1.4. **enum** ServiceOrderStateType

**Description:** Possible values for the state of a Service Order

| State        | Description                                                                                                                               |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| acknowledged | A <b>ServiceOrder</b> request has been received and has passed message and basic validations and a <i>Success Response</i> has been sent. |

| State      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rejected   | <p>This state indicates that:</p> <ul style="list-style-type: none"> <li>- Invalid information is provided through the <code>ServiceOrder</code> / <code>ServiceOrderItem</code> request</li> <li>- The request fails to meet validation rules for <code>Service</code> delivery (processing)</li> </ul> <p>If one <code>ServiceOrderItem</code> is <code>rejected</code>, then the entire <code>ServiceOrder</code> request is <code>rejected</code> and a <i>Error Response</i> is sent.</p>                                                                                                                                                                               |
| inProgress | <p>This state indicates that all <code>ServiceOrderItems</code> have successfully passed the validations checks and the scheduled <code>Service</code> delivery/processing has started. The <code>ServiceOrder</code> will be in <code>inProgress</code> state if <i>at least one</i> <code>ServiceOrderItem</code> is in <code>inProgress</code> state</p>                                                                                                                                                                                                                                                                                                                  |
| pending    | <p>This state indicates that a <code>ServiceOrderItem</code> is currently in a waiting stage for an action/activity to be completed before the order-processing can progress further (this may happen also via non-API channel).</p> <p>A <code>pending</code> state can lead into automatic <code>failed</code> of an <code>ServiceOrderItem</code>, if no action is taken within the agreed timeframe.</p> <p>The <code>ServiceOrder</code> will be in <code>pending</code> state if <i>at least one</i> <code>ServiceOrderItem</code> is in <code>pending</code> state</p>                                                                                                |
| held       | <p>This state indicates that a <code>ServiceOrderItem</code> cannot be progressed due to an issue. The <code>Service</code> delivery (processing) has been temporarily delayed to resolve an infrastructure shortfall to facilitate supply of order. Upon resolution of the issue, the <code>ServiceOrderItem</code> will continue to progress.</p> <p>A <code>held</code> state can lead into automatic <code>failed</code> of an <code>ServiceOrderItem</code>, if no action is taken within the agreed timeframe.</p> <p>The <code>ServiceOrder</code> will be in <code>held</code> state if at least one <code>ServiceOrderItem</code> is in <code>held</code> state</p> |
| failed     | <p>This state indicates that <code>Service</code> delivery (processing) associated with a <code>ServiceOrderItem</code> has failed. This indicates an irrecoverable error as opposed to <code>held</code> or <code>pending</code> issues.</p> <p>The <code>ServiceOrder</code> will be in <code>failed</code> state if <i>ALL</i> <code>ServiceOrderItems</code> are in <code>failed</code> state</p>                                                                                                                                                                                                                                                                        |
| completed  | <p>This state indicates that <code>Service</code> delivery (processing) associated with a <code>ServiceOrderItem</code> has completed.</p> <p>The <code>ServiceOrder</code> will be in <code>completed</code> state if <i>ALL</i> <code>ServiceOrderItems</code> are in <code>completed</code> state</p>                                                                                                                                                                                                                                                                                                                                                                     |
| partial    | <p>This state indicates that some <code>ServiceOrderItem</code> are in <code>completed</code> state while others are in <code>failed</code> states, so the entire <code>ServiceOrder</code> is in a <code>partial</code> state. Not applicable to <code>ServiceOrderItem</code>.</p>                                                                                                                                                                                                                                                                                                                                                                                         |

#### 7.2.1.5. Type ServiceOrderRef

**Description:** Reference to a Service Order instance.

| Name | Type   | M/O | Description                      |
|------|--------|-----|----------------------------------|
| href | string | O   | A hyperlink to the related order |
| id   | string | M   | The id of the related order      |

#### 7.2.1.6. Type ServiceOrderRelationship

**Description:** Reference to a related Service Order and the type of that association.

| Name             | Type                            | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serviceOrder     | <a href="#">ServiceOrderRef</a> | M   | A reference to a Service Order                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| relationshipType | string                          | M   | Specifies the type (nature) of the relationship to the related Service. The nature of required relationships varies for Services of different types. For example, a UNI or ENNI Service may not have any relationships, but an Access E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Services such as multipoint IP or Firewall Services may have more complex relationships. As a result, the allowed and mandatory `relationshipType` values are defined in the Service Specification. |

## 7.2.2. Service Order Item

### 7.2.2.1 Type ServiceOrderItem\_Common

**Description:** An identified part of the order. A service order is decomposed into one or more order items. This type holds the attributes common to request and response representation of the Service Order Item.

| Name                      | Type                                         | M/O | Description                                                                                                                                                              |
|---------------------------|----------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| id                        | string                                       | M   | Identifier of the order item (generally it is a sequence number 01, 02, 03, ...)                                                                                         |
| action                    | <a href="#">ServiceActionType</a>            | M   | Action to be applied to the Service referred by this Service Order Item                                                                                                  |
| coordinatedAction         | <a href="#">OrderItemCoordinatedAction[]</a> | O   | The interval after the completion of one or more related Service Order Items that this Service Order Item can be started or completed                                    |
| note                      | <a href="#">Note_BusSof[]</a>                | O   | Extra-information about the order item; e.g. useful to add extra delivery information that could be useful for a human process                                           |
| relatedContactInformation | <a href="#">RelatedContactInformation[]</a>  | O   | Contact information of an individual or organization playing a role for this Service Order. For providing Notification Contact, `role=notificationContact` MUST be used. |
| service                   | <a href="#">ServiceValue</a>                 | M   | A description of the service that is the subject of this service order item.                                                                                             |

| Name                         | Type                                           | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------|------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| serviceOrderItemRelationship | <a href="#">ServiceOrderItemRelationship[]</a> | O   | Specifies the type (nature) of the relationship to the related Service. The nature of required relationships varies for Services of different types. For example, a UNI or ENNI Service may not have any relationships, but an E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Services such as multipoint IP or Firewall Services may have more complex relationships. As a result, the allowed and mandatory 'relationshipType' values are defined in the Service Specification. Related items can be both from within the same Service Order or from other one. When referencing item within the same Service Order, |

#### 7.2.2.2. Type ServiceOrderItem\_Create

**Description:** An identified part of the order. A service order is decomposed into one or more order items. This type is used in the request.

Inherits from:

- [ServiceOrderItem\\_Common](#)

#### 7.2.2.3. Type ServiceOrderItem

**Description:** An identified part of the order. A service order is decomposed into one or more order items. The modelling pattern introduces the **Common** supertype to aggregate attributes that are common to both **ServiceOrderItem** and **ServiceOrderItem\_Create**. The **Create** type has a subset of attributes of the response type and does not introduce any new, thus the **Create** type has an empty definition

Inherits from:

- [ServiceOrderItem\\_Common](#)

| Name             | Type                                      | M/O | Description                                                                                 |
|------------------|-------------------------------------------|-----|---------------------------------------------------------------------------------------------|
| state            | <a href="#">ServiceOrderItemStateType</a> | M   | State of the Service Order Item                                                             |
| terminationError | <a href="#">TerminationError[]</a>        | O   | When the SOF cannot process the request, the SOF returns a text-based list of reasons here. |

#### 7.2.2.4. **enum** ServiceActionType

**Description:** Action to be applied to the Service referred by this Service Order Item

| ServiceActionType | description |
|-------------------|-------------|
|-------------------|-------------|



| ServiceActionType | description                            |
|-------------------|----------------------------------------|
| add               | Used to create a new Service           |
| modify            | Used to change an existing Service     |
| delete            | Used to disconnect an existing Service |

#### 7.2.2.5. Type ServiceOrderItemRef

**Description:** A reference to a Service Order Item. When referencing item from within the same Service Order, the `serviceOrderId` and `serviceOrderHref` MUST be empty.

| Name             | Type   | M/O | Description                                                       |
|------------------|--------|-----|-------------------------------------------------------------------|
| itemId           | string | M   | Identifier of referenced item within the referenced Service Order |
| serviceOrderHref | string | O   | Link to the order to which the referenced item belongs to         |
| serviceOrderId   | string | O   | Identifier of the order to which the referenced item belongs to   |

#### 7.2.2.6. Type ServiceOrderItemRelationship

**Description:** Specifies the type (nature) of the relationship to the related Service. The nature of required relationships varies for Services of different types. For example, a UNI or ENNI Service may not have any relationships, but an E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Services such as multipoint IP or Firewall Services may have more complex relationships. As a result, the allowed and mandatory `relationshipType` values are defined in the Service Specification. Related item can be both from within the same Service Order or from other one. When referencing item from within the same Service Order, the `orderItem.serviceOrderId` and `orderItem.serviceOrderHref` MUST be empty.

| Name             | Type                                | M/O | Description                                                                                                                                                        |
|------------------|-------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| orderItem        | <a href="#">ServiceOrderItemRef</a> | M   | A reference to a Service Order Item                                                                                                                                |
| relationshipType | string                              | M   | Specifies the nature of the relationship to the related Service Order Item. A string that is one of the relationship types specified in the Service Specification. |

#### 7.2.2.7. `enum` ServiceOrderItemStateType

**Description:** Possible values for the state of a Service Order

| State        | Description                                                                                                                                     |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| acknowledged | A <code>ServiceOrder</code> request has been received and has passed message and basic validations and a <i>Success Response</i> has been sent. |

| State      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rejected   | <p>This state indicates that:</p> <ul style="list-style-type: none"> <li>- Invalid information is provided through the <b>ServiceOrderItem</b> request</li> <li>- The request fails to meet validation rules for <b>Service</b> delivery (processing)</li> </ul> <p>If one <b>ServiceOrderItem</b> is rejected, then the entire <b>ServiceOrder</b> request is rejected and a <i>Error Response</i> is sent.</p>                                                                                                                                                                                                 |
| inProgress | <p>This state indicates that <b>ServiceOrderItem</b> have successfully passed the validations checks and the scheduled <b>Service</b> delivery/processing has started.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| pending    | <p>This state indicates that a <b>ServiceOrderItem</b> is currently in a waiting stage for an action/activity to be completed before the order-processing can progress further (this may happen also via non-API channel).</p> <p>A <i>pending</i> state can lead into automatic <b>failed</b> of an <b>ServiceOrderItem</b>, if no action is taken within the agreed timeframe.</p> <p>The <b>ServiceOrder</b> will be in <i>pending</i> state if <i>at least one</i> <b>ServiceOrderItem</b> is in <i>pending</i> state</p>                                                                                    |
| held       | <p>This state indicates that a <b>ServiceOrderItem</b> cannot be progressed due to an issue. The <b>Service</b> delivery (processing) has been temporarily delayed to resolve an infrastructure shortfall to facilitate supply of order. Upon resolution of the issue, the <b>ServiceOrderItem</b> will continue to progress.</p> <p>A <i>held</i> state can lead into automatic <b>failed</b> of an <b>ServiceOrderItem</b>, if no action is taken within the agreed timeframe.</p> <p>The <b>ServiceOrder</b> will be in <i>held</i> state if at least one <b>ServiceOrderItem</b> is in <i>held</i> state</p> |
| failed     | <p>This state indicates that <b>Service</b> delivery (processing) associated with a <b>ServiceOrderItem</b> has failed. This indicates an irrecoverable error as opposed to <i>held</i> or <i>pending</i> issues.</p> <p>The <b>ServiceOrder</b> will be in <i>failed</i> state if at <i>ALL</i> <b>ServiceOrderItems</b> are in <i>failed</i> state</p>                                                                                                                                                                                                                                                         |
| completed  | <p>This state indicates that <b>Service</b> delivery (processing) associated with a <b>ServiceOrderItem</b> has completed.</p> <p>The <b>ServiceOrder</b> will be in <i>completed</i> state if at <i>ALL</i> <b>ServiceOrderItems</b> are in <i>completed</i> state</p>                                                                                                                                                                                                                                                                                                                                          |

### 7.2.3. Service representation

#### 7.2.3.1. Type ServiceValue

**Description:** ServiceValue is a base class for defining the Service.

| Name        | Type                                   | M/O | Description                                     |
|-------------|----------------------------------------|-----|-------------------------------------------------|
| href        | string                                 | O   | Hyperlink reference to a Service                |
| id          | string                                 | O   | unique identifier of a Service                  |
| description | string                                 | O   | Free-text description of the service            |
| externalId  | string                                 | O   | ID given by the consumer to facilitate searches |
| startDate   | date-time<br><i>format = date-time</i> | O   | Date when the service starts                    |
| endDate     | date-time<br><i>format = date-time</i> | O   | Date when the service ends                      |

| Name                          | Type                                             | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|--------------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| state                         | <a href="#">ServiceStateType</a>                 | O   | Represent the state of lifecycle of the Service Order.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| note                          | <a href="#">Note_BusSof[]</a>                    | O   | A list of notes made on this service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| serviceType                   | string                                           | O   | Business type of the service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| name                          | string                                           | O   | Name of the service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| service<br>Relationship       | <a href="#">Service<br/>Relationship[]</a>       | O   | Specifies the type (nature) of the relationship to the related Service. The nature of required relationships varies for Services of different types. For example, a UNI or ENNI Service may not have any relationships, but an Access E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Services such as multipoint IP or Firewall Services may have more complex relationships. As a result, the allowed and mandatory 'relationshipType' values are defined in the Service Specification. |
| related<br>ContactInformation | <a href="#">Related<br/>ContactInformation[]</a> | O   | Contact information of an individual or organization playing a role for this Service                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| place                         | <a href="#">RelatedPlace<br/>RefOrQuery[]</a>    | O   | The relationships between this Service Order Item and one or more Places as defined in the Service Specification.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| service<br>Configuration      | <a href="#">MefService<br/>Configuration</a>     | O   | MEFServiceConfiguration is used to specify the MEF specific service payload. This field MUST be populated for all item 'actions' other than 'delete'. It MUST NOT be populated when an item 'action' is 'delete'. The @type is used as a discriminator.                                                                                                                                                                                                                                                                                                       |
| serviceOrderItem              | <a href="#">ServiceOrder<br/>ItemRef[]</a>       | O   | A list of service order items related to this service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

### 7.2.3.2. Type MefServiceConfiguration

**Description:** MEFServiceConfiguration is used as an extension point for MEF specific service payload. The @type attribute is used as a discriminator

| Name  | Type   | M/O | Description                                                           |
|-------|--------|-----|-----------------------------------------------------------------------|
| @type | string | M   | The value of the "\$id" as defined in the JSON schema of the service. |

### 7.2.3.3. Type ServiceRelationship

**Description:** A relationship to an existing Service. The requirements for usage for given Service are described in the Service Specification.

| Name             | Type                       | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------|----------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| relationshipType | string                     | M   | Specifies the type (nature) of the relationship to the related Service. The nature of required relationships varies for Services of different types. For example, a UNI or ENNI Service may not have any relationships, but an Access E-Line may have two mandatory relationships (related to the UNI on one end and the ENNI on the other). More complex Services such as multipoint IP or Firewall Services may have more complex relationships. As a result, the allowed and mandatory `relationshipType` values are defined in the Service Specification. |
| service          | <a href="#">ServiceRef</a> | M   | A reference to a Service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

#### 7.2.3.4. **enum ServiceStateType**

**Description:** Valid values for the lifecycle state of the Service.

| State              | Description                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|
| feasibilityChecked | Initial check whether the necessary resources are available and sufficient for the installation of a given service.   |
| designed           | The Service is designed. The resources are identified and/or allocated, but not reserved.                             |
| reserved           | All required resources for given service are reserved and ready.                                                      |
| inactive           | The service is deactivated and is no longer available.                                                                |
| active             | The service is fully available and active                                                                             |
| terminated         | The service is 'logically deleted'. All associated resources are freed and made available for service to other users. |

#### 7.2.3.5. **Type ServiceRef**

**Description:** Reference to a Service instance.

| Name | Type   | M/O | Description                    |
|------|--------|-----|--------------------------------|
| href | string | O   | Hyperlink reference to Service |
| id   | string | M   | unique identifier of Service   |

### 7.2.4. **Place representation**

#### 7.2.4.1. **Type RelatedPlaceRefOrQuery**

**Description:** Allows pointing to a place by referring to a GeographicAddress, GeographicSite, or providing GeographicAddress by value. It also provides additional information like the **role** the place plays for given Product and **contact** needed access to this place.

| Name  | Type                            | M/O | Description |
|-------|---------------------------------|-----|-------------|
| place | <a href="#">PlaceRefOrQuery</a> | M   |             |

| Name    | Type                                 | M/O | Description                                                                                                                 |
|---------|--------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------|
| role    | string                               | M   | Role of this place. The values that can be specified here are described by Product Specification (e.g. "INSTALL_LOCATION"). |
| contact | <a href="#">ContactInformation[]</a> | O   | The person to call to get access to this place in case such access is required to complete the evaluation of this POQ Item. |

#### 7.2.4.2. Type PlaceRefOrQuery

**Description:** A place described by reference to a Geographic Address, Geographic Site or by Geographic Address Representations.

#### 7.2.4.3. Type GeographicAddress\_Query

**Description:** A list of representations being a subset of Geographic Address entity. This is to be used when providing a list of representations to validate a Geographic Address

| Name                            | Type                                              | M/O | Description                                                       |
|---------------------------------|---------------------------------------------------|-----|-------------------------------------------------------------------|
| fieldedAddress Representation   | <a href="#">FieldedAddress Representation[]</a>   | O   | A list of Fielded Address representations                         |
| formattedAddress Representation | <a href="#">FormattedAddress Representation[]</a> | O   | A list of Formatted Address representations                       |
| geographicPoint Representation  | <a href="#">GeographicPoint Representation[]</a>  | O   | A list of Geographic Point Address representations                |
| label Representation            | <a href="#">Label Representation[]</a>            | O   | A list of Label Address representations                           |
| @type                           | string                                            | M   | Used to unambiguously designate the class type when using `oneOf` |

#### 7.2.4.4. Type FieldedAddressRepresentation

**Description:** A type of Address that has a discrete field and value for each type of boundary or identifier down to the lowest level of detail. For example "street number" is one field, "street name" is another field, etc.

| Name               | Type   | M/O | Description                                                                                                           |
|--------------------|--------|-----|-----------------------------------------------------------------------------------------------------------------------|
| streetNr           | string | O   | Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses. |
| streetNrSuffix     | string | O   | The first street number suffix (in a street number range) or the suffix for the street number if there is no range    |
| streetNrLast       | string | O   | Last number in a range of street numbers allocated to an Address                                                      |
| streetNrLastSuffix | string | O   | Last street number suffix for a ranged Address                                                                        |
| streetPreDirection | string | O   | The direction of the street that appears before the Street Name                                                       |

| Name                | Type                                                   | M/O | Description                                                                                                                         |
|---------------------|--------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------|
| streetName          | string                                                 | O   | Name of the street or other street type                                                                                             |
| streetType          | string                                                 | O   | The type of street (e.g., alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf) |
| streetPostDirection | string                                                 | O   | A modifier denoting a relative direction that appears after the Street Name.                                                        |
| poBox               | string                                                 | O   | Number identifying a specific location in a post office.                                                                            |
| locality            | string                                                 | O   | An area of defined or undefined boundaries within a local authority or other legislatively defined area.                            |
| city                | string                                                 | O   | City in which the Address is located.                                                                                               |
| postcode            | string                                                 | O   | A descriptor for a postal delivery area used to speed and simplify the delivery of mail (also known as zip code)                    |
| postcodeExtension   | string                                                 | O   | The extension used on a postal code. Note: there are different use codes for this attribute depending upon the country.             |
| stateOrProvince     | string                                                 | O   | The State or Province in which the Address is located.                                                                              |
| countryCode         | string<br><i>minLength = 2</i><br><i>maxLength = 2</i> | O   | Country in which the Address is located, defined using two characters as defined in ISO 3166                                        |
| subUnit             | SubUnit[]                                              | O   | The Sub Unit represented as a list. This is a list to allow complex sub-unit information such as SUITE 42 ROOM A                    |
| buildingName        | string                                                 | O   | The well-known name of a building that is located at this Address (e.g., where there is one Address for a campus).                  |
| privateStreetNumber | string                                                 | O   | Street number on a private street within the Address.                                                                               |
| privateStreetName   | string                                                 | O   | Private streets internal to a property (e.g., a university) may have internal names that are not recorded by the land title office. |
| language            | string<br><i>minLength = 2</i><br><i>maxLength = 2</i> | O   | The language in which the address is expressed. It MUST use the ISO 639:2023 two letter code 639:2023                               |

#### 7.2.4.5. Type FormattedAddressRepresentation

**Description:** A freeform text representation agreed to by the BUS and SOF.

| Name             | Type   | M/O | Description                                                             |
|------------------|--------|-----|-------------------------------------------------------------------------|
| formattedAddress | string | M   | A formatted Address Representation that contains a non-fielded address. |

| Name     | Type                                                   | M/O | Description                                                              |
|----------|--------------------------------------------------------|-----|--------------------------------------------------------------------------|
| language | string<br><i>minLength = 2</i><br><i>maxLength = 2</i> | O   | The language in which the address is expressed.<br>Based on ISO 639:2023 |

#### 7.2.4.6. Type GeographicPointRepresentation

**Description:** A GeographicPointRepresentation defines a geographic point through coordinates.

| Name       | Type   | M/O | Description                                                                                                                                                 |
|------------|--------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spatialRef | string | M   | The spatial reference system used to determine the coordinates. The system used and the value of this field are to be agreed during the onboarding process. |
| latitude   | string | M   | The latitude expressed in the format specified by the `spacialRef`                                                                                          |
| longitude  | string | M   | The longitude expressed in the format specified by the `spacialRef`                                                                                         |
| elevation  | string | O   | The elevation expressed in the format specified by the `spacialRef`                                                                                         |

#### 7.2.4.7. Type LabelRepresentation

**Description:** A unique identifier controlled by a generally accepted independent administrative authority that specifies a fixed geographical location.

| Name                    | Type                                                   | M/O | Description                                                                                                                                                  |
|-------------------------|--------------------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| label                   | string                                                 | M   | The unique reference to a Geographic Address assigned by the Administrative Authority.                                                                       |
| administrativeAuthority | string                                                 | M   | The organization or standard from the organization that administers this Geographic Address Label ensuring it is unique within the Administrative Authority. |
| language                | string<br><i>minLength = 2</i><br><i>maxLength = 2</i> | O   | The language in which the label is expressed.<br>Based on ISO 639:2023                                                                                       |

#### 7.2.4.8. Type GeographicAddressRef

**Description:** A reference to a Geographic Address resource available through Address Validation API.

| Name | Type   | M/O | Description                                                                                                                                                              |
|------|--------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| href | string | O   | Hyperlink to the referenced Address. Hyperlink MAY be used by the SOF in responses. Hyperlink MUST be ignored by the SOF in case it is provided by the BUS in a request. |
| id   | string | M   | Identifier of the referenced Geographic Address. This identifier is assigned during a successful address validation request (Geographic Address Management API)          |

| Name  | Type   | M/O | Description                                                       |
|-------|--------|-----|-------------------------------------------------------------------|
| @type | string | M   | Used to unambiguously designate the class type when using `oneOf` |

#### 7.2.4.9. Type GeographicSiteRef

**Description:** A reference to a Geographic Site resource available through Service Site API

| Name  | Type   | M/O | Description                                                                                                                                                           |
|-------|--------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| href  | string | O   | Hyperlink to the referenced Site. Hyperlink MAY be used by the SOF in responses. Hyperlink MUST be ignored by the SOF in case it is provided by the BUS in a request. |
| id    | string | M   | Identifier of the referenced Geographic Site.                                                                                                                         |
| @type | string | M   | Used to unambiguously designate the class type when using `oneOf`                                                                                                     |

#### 7.2.4.10. Type SubUnit

**Description:** Allows for sub unit identification

| Name          | Type   | M/O | Description                                                                                 |
|---------------|--------|-----|---------------------------------------------------------------------------------------------|
| subUnitNumber | string | M   | The discriminator used for the subunit, often just a simple number but may also be a range. |
| subUnitType   | string | M   | The type of subunit e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.                |

### 7.2.5. Notification registration

Notification registration and management are done through [/hub](#) API endpoint. The below sections describe data models related to this endpoint.

#### 7.2.5.1. Type EventSubscriptionInput

**Description:** This class is used to register for Notifications.

| Name     | Type   | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------|--------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| callback | string | M   | This callback value must be set to <code>*host*</code> property from Service Order Notification API (serviceOrderNotification.api.yaml). This property is appended with the base path and notification resource path specified in that API to construct an URL to which notification is sent. E.g. for "callback": "https://client.mef.com/listenerEndpoint", the service order state change event notification will be sent to: <code>https://client.mef.com/listenerEndpoint/mefApi/legato/serviceOrderingManagement/v6/listener/serviceOrderStateChangeEvent`</code> |



| Name  | Type   | M/O | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|--------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| query | string | O   | This attribute is used to define to which type of events to register to. Example: "query": "eventType = serviceOrderStateChangeEvent". To subscribe for more than one event type, put the values separated by comma: 'eventType=serviceOrderStateChangeEvent, serviceOrderItemStateChangeEvent'. The possible values are enumerated by 'serviceOrderEventType' in serviceOrderNotification.api.yaml. An empty query is treated as specifying no filters - ending in subscription for all event types. |

### 7.2.5.2. Type EventSubscription

**Description:** This resource is used to respond to notification subscriptions.

| Name     | Type   | M/O | Description                                                                         |
|----------|--------|-----|-------------------------------------------------------------------------------------|
| callback | string | M   | The value provided by in 'EventSubscriptionInput' during notification registration  |
| id       | string | M   | An identifier of this Event Subscription assigned when a resource is created.       |
| query    | string | O   | The value provided by the 'EventSubscriptionInput' during notification registration |

### 7.2.6. Common

Types described in this subsection are shared among two or more LSO APIs.

#### 7.2.6.1. **enum** BusSoftType

**Description:** An enumeration with BUS and SOF values.

#### 7.2.6.2. Type ContactInformation

**Description:** Contact data for a person or organization that is involved in the product offering qualification. In a given context it is always specified by the SOF (e.g. SOF Contact Information) or by the BUS.

| Name            | Type                                         | M/O | Description                                                            |
|-----------------|----------------------------------------------|-----|------------------------------------------------------------------------|
| number          | string                                       | M   | Phone number                                                           |
| emailAddress    | string                                       | M   | Email address                                                          |
| postalAddress   | <a href="#">FieldedAddressRepresentation</a> | O   | Identifies the postal address of the person or office to be contacted. |
| organization    | string                                       | O   | The organization or company that the contact belongs to                |
| name            | string                                       | M   | Name of the contact                                                    |
| numberExtension | string                                       | O   | Phone number extension                                                 |

### 7.2.6.3. Type Duration

**Description:** A Duration in a given unit of time e.g. 3 hours, or 5 days.

| Name   | Type                          | M/O | Description                                        |
|--------|-------------------------------|-----|----------------------------------------------------|
| amount | integer<br><i>minimum = 0</i> | M   | Duration (number of seconds, minutes, hours, etc.) |
| units  | TimeUnit                      | M   | Time unit enumerated                               |

### 7.2.6.4. Type Note\_BusSof

**Description:** Extra information about a given entity. Only useful in processes involving human interaction. Not applicable for an automated process.

| Name   | Type                                   | M/O | Description                                                                                                                   |
|--------|----------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------|
| author | string                                 | M   | Author of the note                                                                                                            |
| date   | date-time<br><i>format = date-time</i> | M   | Date of the note                                                                                                              |
| id     | string                                 | M   | Identifier of the note within its containing entity (may or may not be globally unique, depending on provider implementation) |
| source | BusSofType                             | M   | Indicates if this Note was added by BUS or SOF.                                                                               |
| text   | string                                 | M   | Text of the note                                                                                                              |

### 7.2.6.5. Type OrderCoordinatedAction

**Description:** The interval after the completion of one or more related Order that this Order can be started or completed

| Name                   | Type                                | M/O | Description                                                     |
|------------------------|-------------------------------------|-----|-----------------------------------------------------------------|
| coordinatedActionDelay | Duration                            | M   | The period of time for which the coordinated action is delayed. |
| coordinationDependency | OrderItemCoordinationDependencyType | M   | A dependency between the Order and a related Order              |
| orderId                | string                              | M   | Specifies Order that is to be coordinated with this Order.      |

### 7.2.6.6. Type OrderItemCoordinatedAction

**Description:** The interval after the completion of one or more related Order Items that this Order Item can be started or completed

| Name                   | Type                                | M/O | Description                                                     |
|------------------------|-------------------------------------|-----|-----------------------------------------------------------------|
| coordinatedActionDelay | Duration                            | M   | The period of time for which the coordinated action is delayed. |
| coordinationDependency | OrderItemCoordinationDependencyType | M   | A dependency between the Order Item and a related Order Item    |

| Name   | Type   | M/O | Description                                                          |
|--------|--------|-----|----------------------------------------------------------------------|
| itemId | string | M   | Specifies Order Item that is to be coordinated with this Order Item. |

#### 7.2.6.7. **enum** OrderItemCoordinationDependencyType

**Description:** Possible values of the Order Item Coordination Dependency

| OrderItemCoordinationDependencyType | Description                                                                                        |
|-------------------------------------|----------------------------------------------------------------------------------------------------|
| startToStart                        | Work on the Specified Order Item can only be started after the Coordinated Order Items are started |
| startToFinish                       | The Coordinated Order Items must complete before work on the Specified Order Item begins           |
| finishToStart                       | Work on the Related Order Items begins after the completion of the Specified Order Item            |
| finishToFinish                      | Work on the Related Order Items completes at the same time as the Specified Order Item             |

#### 7.2.6.8. Type RelatedContactInformation

**Description:** Contact information of an individual or organization playing a role for this Order Item. The rule for mapping a represented attribute value to a **role** is to use the *lowerCamelCase* pattern. In a given context it is always specified by the SOF (e.g. SOF Contact Information) or by the BUS.

| Name            | Type                         | M/O | Description                                                            |
|-----------------|------------------------------|-----|------------------------------------------------------------------------|
| role            | string                       | M   | The role of the particular contact in the request                      |
| number          | string                       | M   | Phone number                                                           |
| emailAddress    | string                       | M   | Email address                                                          |
| postalAddress   | FieldedAddressRepresentation | O   | Identifies the postal address of the person or office to be contacted. |
| organization    | string                       | O   | The organization or company that the contact belongs to                |
| name            | string                       | M   | Name of the contact                                                    |
| numberExtension | string                       | O   | Phone number extension                                                 |

The **role** attribute is used to provide a reason the particular party information is used. It can result from business requirements (e.g. SOF Contact Information) or from the Service Specification requirements.

The rule for mapping a represented attribute value to a **role** is to use the *lowerCamelCase* pattern e.g.

- BUS Contact: **role** equal to **busInformation**
- SOF Contact: **role** equal to **sellerContact**

### 7.2.6.9. Type TerminationError

**Description:** This indicates an error that caused an Item to be terminated. The code and propertyPath should be used like in Error422.

| Name         | Type         | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| code         | Error422Code | One of the following error codes: - missingProperty: The property the SOF has expected is not present in the payload - invalidValue: The property has an incorrect value - invalidFormat: The property value does not comply with the expected value format - referenceNotFound: The object referenced by the property cannot be identified in the SOF system - unexpectedProperty: Additional property, not expected by the SOF has been provided - tooManyRecords: the number of records to be provided in the response exceeds the SOF's threshold. - otherIssue: Other problem was identified (detailed information provided in a reason) |
| propertyPath | string       | A pointer to a particular property of the payload that caused the validation issue. It is highly recommended that this property should be used. Defined using JavaScript Object Notation (JSON) Pointer ( <a href="https://tools.ietf.org/html/rfc6901">https://tools.ietf.org/html/rfc6901</a> ).                                                                                                                                                                                                                                                                                                                                            |
| value        | string       | Text to describe the reason of the termination.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

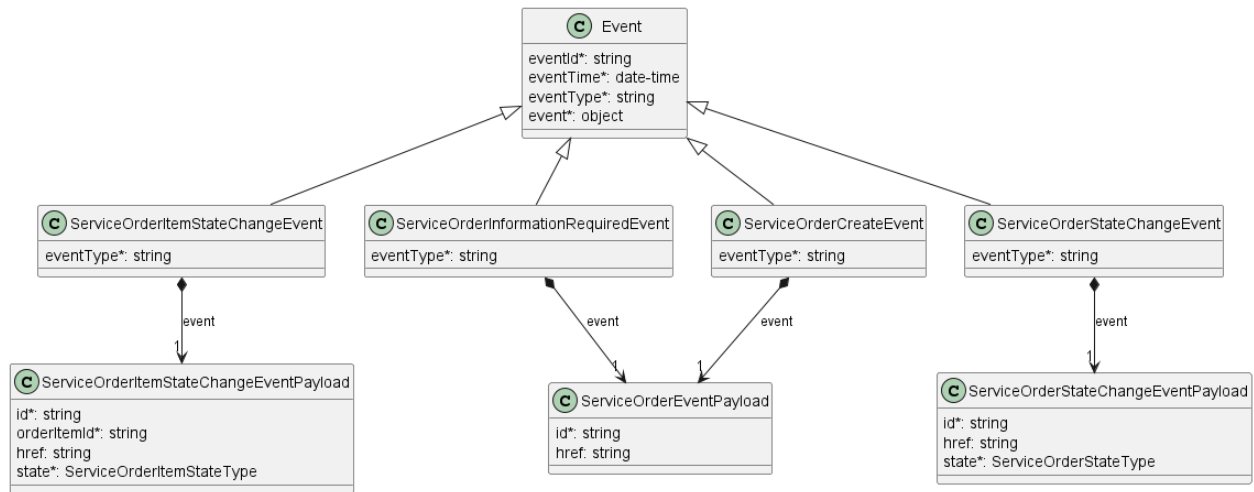
### 7.2.6.10. enum TimeUnit

**Description:** Represents a unit of time.

| Value         |
|---------------|
| seconds       |
| minutes       |
| businessHours |
| calendarHours |
| businessDays  |
| calendarDays  |
| months        |
| years         |

## 7.3. Notification API Data model

Figure 27 presents the Service Order Management Notification data model.



**Figure 27. Service Order Management Notification Data Model**

This data model is used to construct requests and responses of the API endpoints described in [Section 5.2.2](#).

### 7.3.1. Type Event

**Description:** Event class is used to describe information structure used for notification.

| Name      | Type                                   | M/O | Description                                      |
|-----------|----------------------------------------|-----|--------------------------------------------------|
| eventId   | string                                 | M   | Id of the event                                  |
| eventTime | date-time<br><i>format = date-time</i> | M   | Date-time when the event occurred                |
| eventType | string                                 | M   | The type of the notification.                    |
| event     | object                                 | M   | The event linked to the involved resource object |

### 7.3.2. Type ServiceOrderCreateEvent

**Description:**

Inherits from:

- [Event](#)

| Name      | Type                                     | M/O | Description                                                   |
|-----------|------------------------------------------|-----|---------------------------------------------------------------|
| eventType | string                                   | M   | Indicates the type of the event.                              |
| event     | <a href="#">ServiceOrderEventPayload</a> | M   | A reference to the object that is source of the notification. |

### 7.3.3. Type ServiceOrderEventPayload

**Description:** The identifier of the Service Order and Order Item being subject of this event.

| Name | Type   | M/O | Description                           |
|------|--------|-----|---------------------------------------|
| id   | string | M   | ID of the Service Order               |
| href | string | O   | Hyperlink to access the Service Order |

### 7.3.4. Type ServiceOrderInformationRequiredEvent

#### Description:

Inherits from:

- [Event](#)

| Name      | Type                                     | M/O | Description                                                   |
|-----------|------------------------------------------|-----|---------------------------------------------------------------|
| eventType | string                                   | M   | Indicates the type of the event.                              |
| event     | <a href="#">ServiceOrderEventPayload</a> | M   | A reference to the object that is source of the notification. |

### 7.3.1. Type ServiceOrderItemStateChangeEvent

#### Description:

Inherits from:

- [Event](#)

| Name      | Type                                                    | M/O | Description                                                   |
|-----------|---------------------------------------------------------|-----|---------------------------------------------------------------|
| eventType | string                                                  | M   | Indicates the type of the event.                              |
| event     | <a href="#">ServiceOrderItemStateChangeEventPayload</a> | M   | A reference to the object that is source of the notification. |

### 7.3.2. Type ServiceOrderItemStateChangeEventPayload

**Description:** The identifier of the ServiceOrderItem being subject of this event.

| Name    | Type                                      | M/O | Description                                                                                                                                      |
|---------|-------------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------|
| id      | string                                    | M   | ID of the ServiceOrder                                                                                                                           |
| orderId | string                                    | M   | ID of the Service Order Item (within the Service Order) which state change triggered the event. Mandatory for `serviceOrderItemStateChangeEvent` |
| href    | string                                    | O   | Hyperlink to access the ServiceOrder                                                                                                             |
| state   | <a href="#">ServiceOrderItemStateType</a> | M   | The state of the Service Order                                                                                                                   |

### 7.3.3. Type ServiceOrderStateChangeEvent

#### Description:

Inherits from:

- [Event](#)

| Name      | Type                                                | M/O | Description                                                   |
|-----------|-----------------------------------------------------|-----|---------------------------------------------------------------|
| eventType | string                                              | M   | Indicates the type of the event.                              |
| event     | <a href="#">ServiceOrderStateChangeEventPayload</a> | M   | A reference to the object that is source of the notification. |

### 7.3.4. Type ServiceOrderStateChangeEventPayload

**Description:** The identifier of the ServiceOrder being subject of this event.

| Name  | Type                                       | M/O | Description                          |
|-------|--------------------------------------------|-----|--------------------------------------|
| id    | string                                     | M   | ID of the ServiceOrder               |
| href  | string                                     | O   | Hyperlink to access the ServiceOrder |
| state | <a href="#">ServiceOrder<br/>StateType</a> | M   | The state of the Service Order       |

## 8. References

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## Appendix A Acknowledgments

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